

ChE 301 – Chemical Engineering Thermodynamics –Spring 2026

Student Learning Outcome 1

Rubric for assessment: Assignments #1,#2A,#2B, #3

1. **Assignment #1: Analysis of an Exponential Equation of State.**
2. **Assignment #2A: Thermal Properties of N₂ from Experimental Data**
3. **Assignment #2B: Ideal Gas Venting. Pressure Relief Valve Modeling**
4. **Assignment #3: Liquefaction of Nitrogen via the Linde Process**

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Assignment #1 100 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<60 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (60-80 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (>80 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Assignment #2A, 2B 100 pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<60 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (60-80 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (>80 pts)
4. Translates academic theory into engineering applications Assignment #3 100 pts	Does not appear to grasp the connection between theory and the problem (<60 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (60-80 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (>80 pts)

Rubric					
PI	Problem	Total pts	Unsatisfactory	Developing	Satisfactory
1	1[Assign #1]	100	<60	60-80	>80
2	2[Assign #2A,2B]	100	<60	60-80	>80
4	4[Assign #3]	100	<60	60-80	>80

RESULTS BY PERFORMANCE INDICATOR			
Rubric #	1	2	4
Mapped SLO	1	1	1
Average	2.50	2.67	2.64
Standard Deviation	0.51	0.59	0.67
S	9	13	8
D	9	4	2
U	0	1	1
Checksum	18	18	11

RESULTS BY STUDENT LEARNING OUTCOME			
SLO	1		
Average Score	2.60		
Standard Deviation	0.07		

Analysis of Results

Average scores were fairly uniform although there was some variation among students. More students were strong in solving engineering problems using theoretical thermodynamic concepts , and concepts of algebra, calculus, differential equations and/or linear algebra. Overall performance was satisfactory for all performance indicators. The more stringent limits for unsatisfactory, developing and satisfactory performance reflect the expanded use of AI as well as the issuance of my textbook in Chemical Engineering Thermodynamics.

Assignment #1: Analysis of an Exponential Equation of State.

The following **exponential** Equation of State (EOS) has been proposed as an improvement to the Van der Waals equation.

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right)$$

$$y = \frac{b}{4V} \quad a = \lambda_1 \frac{(RT_c)^2}{P_c} \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Data on the Nitrogen gas compressibility factor were reported by Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952).

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

1. Using the proper thermodynamic and mathematical arguments estimate the numerical values of the constants $\{\lambda_1, \lambda_2\}$ and the critical compressibility factor.
2. Perform a nonlinear regression and calculate the constants $\{\lambda_1, \lambda_2\}$ using the experimental data for Nitrogen given in the table above.
3. Comment with a brief paragraph on whether the values calculated from the nonlinear regression are thermodynamically consistent
4. Derive all the formulas for the calculation of the primary residual thermal properties, fugacity coefficient, enthalpy and entropy.
5. Derive all the formulas for the calculation of the secondary thermal properties in the following order: thermal expansion coefficient, heat capacity under constant volume, heat capacity under constant pressure, Joule-Thomson coefficient and speed of sound.

6. Calculate at temperature 10 °C and pressure 830 bar

1. The molar enthalpy in **kJ/mol**
2. The molar entropy in **kJ/(kmol K)**
3. The thermal expansion coefficient in **K⁻¹**
4. The molar heat capacity C_V in **kJ/(kmol K)**
5. The molar heat capacity C_p in **kJ/(kmol K)**
6. The Joule-Thomson coefficient in **K/bar**
7. The speed of sound in **m/s**

7. Prepare a Table with the following columns

- a. Column 1: Property description (See Excel Template **Assignment-04.xlsm**)
- b. Column 3: Value of property from **Assignment #03**
- c. Column 4: Value of property using the Peng-Robinson Equation of State
- d. Column 5: Value of property using the exponential EOS with theoretical values of $\{\lambda_1, \lambda_2\}$
- e. Column 6: Value of property using the exponential EOS with numerical values of $\{\lambda_1, \lambda_2\}$
- f. Column 7: NIST value from the table below

NIST Data

Isothermal Data for T = 10.000 C										
Temperature (C)	Pressure (bar)	Density (mol/l)	Volume (l/mol)	Internal Energy (kJ/mol)	Enthalpy (kJ/mol)	Entropy (J/mol*K)	Cv (J/mol*K)	Cp (J/mol*K)	Sound Spd. (m/s)	Joule-Thomson (K/bar)
10.000	800.00	19.302	0.051807	6.6840	10.829	48.874	23.750	38.775	764.43	-0.046582
10.000	830.00	19.596	0.051032	6.6479	10.884	48.523	23.821	38.731	778.50	-0.047978
10.000	860.00	19.877	0.050308	6.6135	10.940	48.186	23.891	38.691	792.28	-0.049248
10.000	890.00	20.149	0.049631	6.5806	10.998	47.860	23.960	38.653	805.77	-0.050405
10.000	920.00	20.410	0.048996	6.5492	11.057	47.546	24.029	38.617	819.00	-0.051462
10.000	950.00	20.662	0.048398	6.5192	11.117	47.243	24.096	38.584	831.98	-0.052429
10.000	980.00	20.905	0.047834	6.4904	11.178	46.949	24.163	38.553	844.70	-0.053314
10.000	1010.0	21.141	0.047301	6.4629	11.240	46.665	24.228	38.524	857.19	-0.054126

Auxiliary Data	
Reference States, Normal Boiling Point Convention	
Enthalpy	H = 0 at the normal boiling point for saturated liquid.
Entropy	S = 0 at the normal boiling point for saturated liquid.
Additional fluid properties	
Critical temperature (T _c)	-146.958 C
Critical pressure (P _c)	33.958 bar
Critical density (D _c)	11.1839 mol/l
Acentric factor	0.0372
Normal boiling point	-195.795 C
Dipole moment	0.0 Debye

Assignment #2A: Thermal Properties of Nitrogen from Experimental Data

The following data on $\frac{P}{\rho RT}$ for Nitrogen gas were reported by Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952).

	Temperature (deg C)				
P (atm)	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

Calculate at temperature 10 °C and pressure 830 bar

8. The thermal expansion coefficient in **K⁻¹**
9. The pressure expansion coefficient in **K⁻¹**
10. The molar enthalpy in **kJ/mol**
11. The molar entropy in **kJ/(kmol K)**
12. The molar heat capacity C_p in **kJ/(kmol K)**
13. The molar heat capacity C_v in **kJ/(kmol K)**
14. The Joule-Thomson coefficient in **K/bar**
15. The speed of sound in **m/s**

Hint: Miscellaneous Derivatives needed for the solution

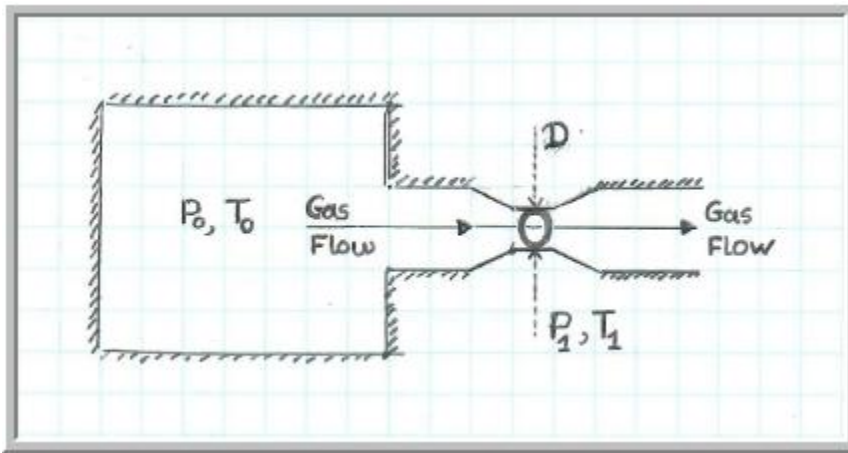
$$\left. \frac{\partial P}{\partial T} \right|_{\rho} = -\frac{\rho_T}{\rho_P} \quad \left. \frac{\partial^2 P}{\partial T^2} \right|_{\rho} = \frac{\rho_{TT}\rho_P^2 - 2\rho_T\rho_{TP}\rho_P + \rho_{PP}\rho_T^2}{\rho_P^3}$$

where

$$\rho_T = \left. \frac{\partial \rho}{\partial T} \right|_P \quad \rho_P = \left. \frac{\partial \rho}{\partial P} \right|_T \quad \rho_{TT} = \left. \frac{\partial^2 \rho}{\partial T^2} \right|_P \quad \rho_{PP} = \left. \frac{\partial^2 \rho}{\partial P^2} \right|_T \quad \rho_{TP} = \left. \frac{\partial^2 \rho}{\partial T \partial P} \right|$$

Assignment #2B: Ideal Gas Venting.

Consider the venting of ethylene gas from the carbon steel tank shown in the diagram below



As the pressure begins to drop the gas velocity increases until it reaches the speed of sound in the narrow section of the pipe. **This is known as the critical condition.** The initial pressure and temperature in the tank are equal to 200 bar and 400 K respectively. The volume of the tank is equal to 20 m³

Calculate:

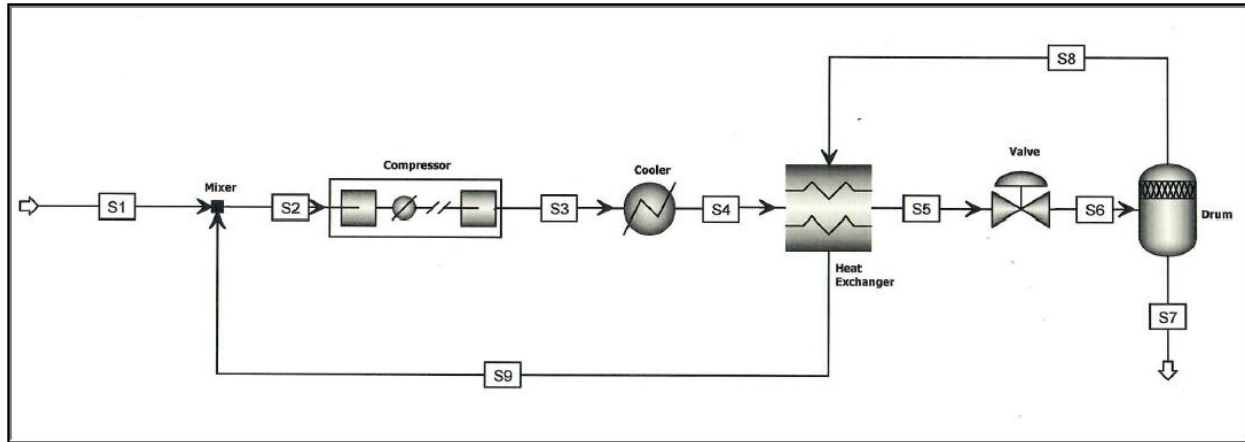
1. The initial mass in **kg** of ethylene in the tank
2. The temperature and pressure at the narrow section of the pipe when the critical mass flow rate which corresponds to the speed of sound (critical condition) is $m=5000$ kg/hr.
3. The diameter in **mm** of the narrow section of the venting pipe
4. The mass in **kg** of ethylene in the tank when the critical condition is reached
5. When during the venting process the pressure gauge reads $P_1=2300$ **psig** what is the mass in **kg** of ethylene in the tank

Assumptions and Hints

1. **Assumption #1:** The venting process is **reversible** and **adiabatic**
2. **Assumption #2:** Ethylene behaves as an ideal gas
3. **Hint #1:** You must start with the unsteady-state mass, energy and entropy balances
4. **Hint #2:** The kinetic energy of the gas at the narrow section of the pipe cannot be ignored.

Assignment #3: Liquefaction of Nitrogen via the Linde Process

Consider the liquefaction of Nitrogen via the [Linde](#) process shown in the process flow diagram(PFD) below. Stream numbers are indicated in the squares.



In class we have analyzed this process and estimated the thermodynamic efficiency to be approximately 4%. This is a very low efficiency resulting in large capital and operating expenses. In order to improve the efficiency we have decided to replace the single stage compressor with a four stage compressor with intercoolers. The intercoolers cool each compressor stage discharge down to 30 °C. Furthermore we have decided to replace the valve with a liquid turbine(expander). The following data are available for this process:

Stream No →	S1	S2	S3	S4	S5	S6	S7	S8	S9
Temperature, °C	20.0			30.0		T^{sat}	T^{sat}	T^{sat}	25.0
Pressure, bar	1.0	1.0	180.0	180.0	180.0	1.0	1.0	1.0	1.0
Flow Rate, MMSCFD	50.0								

For your solution consider the following data and assumptions:

1. Thermodynamic properties of Nitrogen are described by the **Peng-Robinson** equation of state
2. The efficiency of the compressor and the liquid turbine(expander) is 85%
3. No pressure drop in the separator drum, all the coolers and the heat exchanger.

Calculations:

- Prepare a detailed report, outlining your logic and calculations similar to the one we have developed in class.
- Prepare a small Excel table, showing all the balance equations you are using to solve this problem along with the lost work per equipment (actual and percent), the total lost work and the thermodynamic efficiency
- At the end of the report attach an image of excel calculations in a form of a material and energy balance similar to the one we have created in class.

Chemical Engineering Thermodynamics

Assignment #04

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS

Term: Spring 2026

Professor: Dr. Michael Caracotsios

Name: Lama Alsurakhi

Due Date: Friday, March 20th

Abstract

This report analyzes the thermodynamic properties of nitrogen at high pressure using an exponential equation of state and compares the results with experimental data and with the Peng-Robinson equation of state. The work first determines the parameters λ_1 , λ_2 , and Z_c mathematically from critical point conditions and then determines fitted parameter values using Solver. Using these parameter sets, expressions for the fugacity coefficient, enthalpy, entropy, heat capacities, thermal expansion coefficient, Joule-Thomson coefficient, and speed of sound are developed and applied. Results obtained from the exponential EOS with theoretical and experimental parameters are compared with values from Assignment #3 and from the CHE Calculator. The comparison shows that the experimentally fitted parameters provide better agreement with the property data at the specified state, while the theoretical parameters satisfy the critical thermodynamic constraints of the model.

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1 Background

Chemical engineering thermodynamics provides the framework for calculating thermal properties such as enthalpy, entropy, fugacity, heat capacities, and density. In the study of pure fluids, these properties can be developed from thermodynamic identities and from equations of state that relate pressure, temperature, and volume or density.

For real fluids, the ideal-gas model is often not sufficient, especially at high pressures. Because of this, equations of state are used to describe non-ideal behavior. These models allow the calculation of compressibility factor, fugacity coefficient, residual enthalpy, residual entropy, and other secondary properties. In this assignment, both the exponential equation of state and the Peng-Robinson equation of state are used as tools for property prediction and comparison..

The report also uses nitrogen property data such as critical constants and the acentric factor. These values are needed to define EOS parameters and reduced properties, and they were taken from NIST data. In addition, ChatGPT and Gemini were used as supplemental tools for explanation, formatting, and organization support during the preparation of this report.

2 Introduction

The purpose of this assignment is to analyze the thermodynamic behavior of nitrogen at the specified state using multiple methods and to compare the results. The report begins by finding λ_1 , λ_2 , and Z_c mathematically from the critical-point constraints of the exponential EOS. It then uses Solver to determine experimental values of λ_1 and λ_2 that better fit the available compressibility data.

After determining the parameter values, the report develops equations for fugacity coefficient, residual enthalpy, and residual entropy in terms of density. It then extends the analysis to secondary properties including the thermal expansion coefficient β , the molar heat capacities C_p and C_v , the Joule-Thomson coefficient μ_{JT} , and the speed of sound v . These properties are evaluated using experimental data, the Peng-Robinson EOS in the CHE Calculator, and the exponential EOS with both theoretical and experimental parameter values.

The main goal is not only to calculate the properties, but also to compare the strengths and limitations of each method. In this way, the assignment shows how theoretical derivations, fitted models, and reference data can all be used together in thermodynamic analysis.

3 finding λ_1 , λ_2 , and Z_c mathematically:

$$Z = \frac{PV}{RT} = \frac{1+y+y^2-y^3}{(1-y)^3} \exp\left(-\frac{a}{VRT}\right) \quad y = \frac{b}{4V} \quad a = \lambda_1 \frac{(RT_c)^2}{P_c} \quad b = \lambda_2 \frac{RT_c}{P_c}$$

at the Critical Point

$$\left(\frac{\partial P}{\partial V}\right)_T = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$$

So we will solve for P

And solve for V in $y = \frac{b}{4V} \Rightarrow V = \frac{b}{4y}$ and

$$Z = \frac{1+y+y^2-y^3}{(1-y)^3} \exp\left(-\frac{a}{VRT}\right)$$

then

$$P = \frac{ZRT}{V} = \frac{4RTy}{b} \frac{1+y+y^2-y^3}{(1-y)^3} \exp\left(-\frac{4ya}{bRT}\right)$$

And since by definition $T = T_c$ we know

$$\frac{a}{bRT_c} \Rightarrow \frac{\left[\lambda_1 \frac{(RT_c)^2}{P_c}\right]}{\left[\lambda_2 \frac{RT_c}{P_c}\right] RT_c} \Rightarrow \frac{\lambda_1}{\lambda_2}, \quad \text{So,}$$

$$P_c = \frac{4RT_c y_c}{b} \cdot \frac{1+y_c+y_c^2-y_c^3}{(1-y_c)^3} \exp\left(-4\frac{\lambda_1}{\lambda_2} y_c\right), \quad \text{let } \frac{4\lambda_1}{\lambda_2} = g$$

$$P_c = \frac{4RT_c y_c}{b} \cdot \frac{1+y_c+y_c^2-y_c^3}{(1-y_c)^3} \exp(-gy_c) \quad \text{now we only have two unknowns}$$

Now we will solve for y_c by Rewriting

$$P_c = \frac{4RT_c}{b} K(y)$$

where

$$K(y) = y \frac{1+y+y^2-y^3}{(1-y)^3} \exp(-gy_c)$$

And since $y = \frac{b}{4V}$ and $\left(\frac{\partial P}{\partial V}\right)_T = 0$ so $V = \frac{b}{4y}$ will be $\left(\frac{\partial P}{\partial y}\right)_T = 0$

And therefore

$$\frac{\partial K}{\partial y} = 0$$

we will take the log of each side

$$\ln K(y) = \ln y + \ln(1+y+y^2-y^3) - 3\ln(1-y) - gy_c \quad \text{we know that } \frac{\partial \ln K}{\partial y} = 0$$

So,

$$0 = \frac{1}{y} + \frac{1 + 2y_c - 3y_c^2}{1 + y_c + y_c^2 - y_c^3} + \frac{3}{1 - y_c} - g \quad \text{Solve for } g$$

Let's set

$$f(y) = \frac{1}{y_c} + \frac{1 + 2y_c - 3y_c^2}{1 + y_c + y_c^2 - y_c^3} + \frac{3}{1 - y_c}$$

so therefore $g = f(y)$ and since $\frac{\partial^2 P}{\partial y^2} = 0$ then $f'(y) = 0$

$$0 = -\frac{1}{y_c^2} + \frac{(2 - 6y_c)(1 + y_c + y_c^2 - y_c^3) - (1 + 2y_c - 3y_c^2)(1 + 2y_c - 3y_c^2)}{(1 + y_c + y_c^2 - y_c^3)^2} + \frac{3}{(1 - y_c)^2}$$

Now solve for y we'll use goal seek to solve for this

once we do that we get

$$y_c = 0.38213$$

Since $g = f(y)$ we can solve for it using the value of y_c

$$g = \frac{1}{0.38} + \frac{1 + 2(0.38) - 3(0.38)^2}{1 + 0.38 + (0.38)^2 - (0.38)^3} + \frac{3}{1 - 0.38} \quad \text{to give us } g = 8.373$$

Now that we have g and y_c we can find Z_c

$$Z_c = \frac{1 + y_c + y_c^2 - y_c^3}{(1 - y_c)^3} \exp(-gy_c)$$

Plug everything in to get

$$Z_c = 0.2545$$

Now we will start solving for λ_2 we know $b = \lambda_2 \frac{RT_c}{P_c}$ and $y_c = \frac{b}{4V_c}$

We Can solve for $\lambda_2 \Rightarrow \lambda_2 = \frac{bP_c}{RT_c}$ and for $b \Rightarrow b = 4y_c V_c$

and substitute.

$$\lambda_2 = \frac{4y_c V_c P_c}{RT_c} \quad \text{Since } Z_c = \frac{P_c V_c}{RT_c} \text{ we Can substitute again}$$

$$\lambda_2 = 4y_c Z_c$$

we Can solve for λ_2 to give us

$$\lambda_2 = 0.38907$$

Now we Can find λ_1 from $g = \frac{4\lambda_1}{\lambda_2}$ by substituting all values in and

Solving for $\lambda_1 \Rightarrow \lambda_1 = \frac{g\lambda_2}{4}$ to give us

$$\lambda_1 = 0.81442$$

1	
yc	0.382131883
f'(y)	1.01019E-09
g	8.373028782
alpha, g*y	3.199601254
Zc	0.254540144
Λ_1	0.814426965
Λ_2	0.389071619

Figure 1: Excel Set Up to find Z_c and the Theoretical λ

4 Finding λ_1 and λ_2 using Solver

To find the experimental λ s we would have to first find a and b using the λ s we have calculated in problem 1

$$a = 264.1371424 \text{ KJ} \cdot \text{m}^3/\text{kmol}^2$$

$$b = 0.1202645159 \text{ m}^3/\text{kmol}$$

Once we get those we will set up a table that includes the pressure, temperature, the experimental Z, V, y, and Z calculated using the given EOS formula and then error and $error^2$ for the experimental Z and the Z calculated once we have that we will take the sum of it and use solver to minimize the sum by changing the λ s once we do that we get

$$\lambda_1 = 0.26329 \quad \text{and} \quad \lambda_2 = 0.17663$$

19	λ_1	0.263292745		
20	λ_2	0.176634575		
21	R, KJ/Kmol K	8.314		
22	Tc	126.2		
23	Pc	33.943875		
24	a	8539.181086	a in KJ*m^3/Kmol^2	85.39181086
25	b	5.459887224	b in m^3/Kmol	0.054598872
26	KJ to m^3*bar	100		

Figure 2: Solver results for experimental λ_1 and λ_2

P_bar	T_C	T_K	Zexp	Vexp, ZRT/P, m^3Kmol^-1	y, b4V	Zcalc	error	error^2
0	0	273.15	1	0	0	0		
0	25	298.15	1	0	0	0		
0	50	323.15	1	0	0	0		
0	75	348.15	1	0	0	0		
0	100	373.15	1	0	0	0		
1.01325	0	273.15	1	22.41272243	0.000609	1.000759411	0.000759	5.76705E-07
1.01325	25	298.15	1	24.46404244	0.000558	1.000824632	0.000825	6.80018E-07
1.01325	50	323.15	1	26.51536245	0.000515	1.000861358	0.000861	7.41938E-07
1.01325	75	348.15	1	28.56668246	0.000478	1.000879407	0.000879	7.73357E-07
1.01325	100	373.15	1	30.61800247	0.000446	1.000885045	0.000885	7.83305E-07
3.03975	0	273.15	0.999	7.463436568	0.001829	1.002286717	0.003287	1.08025E-05
3.03975	25	298.15	0.999	8.146526132	0.001676	1.002482169	0.003482	1.21255E-05
3.03975	50	323.15	1	8.838454149	0.001544	1.002589488	0.002589	6.70545E-06
3.03975	75	348.15	1	9.522227486	0.001433	1.002643288	0.002643	6.98637E-06
3.03975	100	373.15	1.001	10.21620682	0.001336	1.002657212	0.001657	2.74635E-06
5.06625	0	273.15	0.998	4.473579397	0.003051	1.003825404	0.005825	3.39353E-05
5.06625	25	298.15	0.999	4.887915679	0.002793	1.004146626	0.005147	2.64878E-05
5.06625	50	323.15	1	5.30307249	0.002574	1.004324853	0.004325	1.87044E-05
5.06625	75	348.15	1.001	5.719049828	0.002387	1.00440951	0.00341	1.16248E-05
5.06625	100	373.15	1.001	6.129724094	0.002227	1.004436587	0.003437	1.18101E-05
10.1325	0	273.15	0.996	2.232307154	0.006115	1.007718532	0.011719	0.000137324
10.1325	25	298.15	0.998	2.441511435	0.005591	1.00835027	0.01035	0.000107128
10.1325	50	323.15	1	2.651536245	0.005148	1.008695073	0.008695	7.56043E-05
10.1325	75	348.15	1.001	2.859524914	0.004773	1.008861396	0.007861	6.18016E-05
10.1325	100	373.15	1.002	3.067923847	0.004449	1.008903851	0.006904	4.76632E-05
30.3975	0	273.15	0.959	0.716460027	0.019052	1.024746954	0.065747	0.004322662
30.3975	25	298.15	0.996	0.812206209	0.016806	1.025695773	0.029696	0.000881839
30.3975	50	323.15	1.001	0.88472926	0.015428	1.02660954	0.02561	0.000655849
30.3975	75	348.15	1.005	0.956983862	0.014263	1.026988596	0.021989	0.000483498
30.3975	100	373.15	1.007	1.027744283	0.013281	1.027054061	0.020054	0.000402165
50.6625	0	273.15	0.958	0.429427762	0.031786	1.042459491	0.084459	0.007133406
50.6625	25	298.15	0.996	0.487323725	0.02801	1.043838243	0.047838	0.002288497
50.6625	50	323.15	1.004	0.532428478	0.025637	1.045144963	0.041145	0.001692909
50.6625	75	348.15	1.009	0.576475652	0.023678	1.045661239	0.036661	0.001344046
50.6625	100	373.15	1.014	0.62093309	0.021983	1.045570051	0.03157	0.000996668
101.325	0	273.15	0.984	0.220541189	0.061892	1.088286489	0.104286	0.010875672
101.325	25	298.15	1.004	0.245618986	0.055573	1.092119719	0.08812	0.007765085
101.325	50	323.15	1.018	0.26992639	0.050568	1.093693248	0.075699	0.005730376
101.325	75	348.15	1.027	0.293379829	0.046526	1.093975537	0.066976	0.004485722
101.325	100	373.15	1.033	0.316283965	0.043157	1.093385931	0.060386	0.003646461
202.65	0	273.15	1.036	0.116097902	0.117571	1.189300526	0.153301	0.023501051
202.65	25	298.15	1.057	0.129292464	0.105572	1.194364843	0.137365	0.0188691
202.65	50	323.15	1.072	0.142122343	0.096042	1.195502444	0.123502	0.015252854
202.65	75	348.15	1.082	0.154545752	0.088322	1.194423791	0.112424	0.012639109
202.65	100	373.15	1.089	0.166715023	0.081875	1.191879319	0.102879	0.010584154
303.975	0	273.15	1.132	0.084570673	0.1614	1.286222871	0.154223	0.023784694
303.975	25	298.15	1.146	0.093452642	0.14606	1.293257537	0.147258	0.021684782
303.975	50	323.15	1.154	0.101995761	0.133826	1.295090335	0.14109	0.019906483
303.975	75	348.15	1.16	0.110457839	0.123574	1.293024079	0.133024	0.017695406
303.975	100	373.15	1.163	0.11869579	0.114997	1.289037119	0.126037	0.015885355
405.3	0	273.15	1.256	0.070375948	0.193954	1.370122183	0.114122	0.013023873
405.3	25	298.15	1.254	0.076694773	0.177975	1.383264838	0.129265	0.016709398
405.3	50	323.15	1.253	0.083059373	0.164337	1.387187315	0.134187	0.018006235
405.3	75	348.15	1.25	0.089270883	0.152902	1.386334926	0.136335	0.018587212
405.3	100	373.15	1.247	0.095451623	0.143001	1.381911859	0.134912	0.01820121
607.95	0	273.15	1.524	0.056928315	0.23977	1.509034202	-0.01497	0.000223975
607.95	25	298.15	1.495	0.060956239	0.223927	1.535340206	0.04034	0.001627332
607.95	50	323.15	1.471	0.06500683	0.209974	1.54824143	0.077241	0.005966239
607.95	75	348.15	1.45	0.069036149	0.197718	1.552434438	0.102434	0.010492814
607.95	100	373.15	1.432	0.073074966	0.186791	1.550403291	0.118403	0.014019339
810.6	0	273.15	1.798	0.050372594	0.270975	1.620466768	-0.17753	0.031518049
810.6	25	298.15	1.723	0.052689431	0.25906	1.673306918	-0.04969	0.002469402
810.6	50	323.15	1.697	0.056245713	0.24268	1.684313243	-0.01269	0.000160954
810.6	75	348.15	1.658	0.059204449	0.230552	1.695728066	0.037728	0.001423407
810.6	100	373.15	1.625	0.062192818	0.219474	1.698212903	0.073213	0.005360129
1013.25	0	273.15	2.067	0.046327097	0.294638	1.715876784	-0.35112	0.123287513
1013.25	25	298.15	1.988	0.048634516	0.280659	1.769350448	-0.21865	0.047807613
1013.25	50	323.15	1.923	0.050989042	0.267699	1.80239668	-0.1206	0.014545161
1013.25	75	348.15	1.867	0.053333996	0.255929	1.821503838	-0.0455	0.002069901
1013.25	100	373.15	1.819	0.055694146	0.245084	1.82980021	0.0108	0.000116645

Figure 3: Excel Table Set Up for Problem 2

Sum	0.578699248
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Figure 4: Sum After Solver

5 Comparing Theoretical Vs. Experimental λ

The theoretical values of λ_1 and λ_2 are the values Calculated in question 1 which gave us $\lambda_1 = 0.8144$ and $\lambda_2 = 0.38907$ and the experimental values are the ones Calculated in question 2 which gave us $\lambda_1 = 0.26329$ and $\lambda_2 = 0.17663$

Since the experimental values are much smaller then the actual values they do not Satisfy the critical thermodynamic constraints of the model. Although they give a reasonable fit for the experimental Z data over the Pressure range they are not thermodynamically Consistent.

6 Deriving H^R , S^R , and ϕ

Let's write things in terms of density

$$\rho = \frac{1}{V} \quad \text{so,} \quad y = \frac{b\rho}{4}$$

and let's define the inside of the exponential q just like previously

So

$$q = \frac{a}{RT}$$

so then

$$Z(\rho, T) = K(y) \exp(-q\rho) \quad \text{where} \quad K(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}$$

6.1 Fugacity Coefficient, ϕ

In Chapter 4 it gives us the following formula for fugacity:

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho (Z - 1) \frac{d\rho}{\rho}$$

$$\boxed{\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \left[\frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a\rho}{RT}\right) - 1 \right] \frac{d\rho}{\rho} \quad \text{and} \quad y = \frac{b\rho}{4}}$$

6.2 Molar Enthalpy, H^R

In Chapter 4 we are given the formula for residual enthalpy as follows:

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho \left(T \frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}$$

First we need to find $\left(\frac{\partial Z}{\partial T} \right)_\rho$ to do so we will let

$$P(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}$$

So therefore,

$$Z = P(y) \exp(-q\rho) \quad , \quad \text{where} \quad q = \frac{a}{RT}$$

Now since only the exponential is the only function that depends on T and everything else is Constant we Can take the derivative as follows:

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = P(y) \exp\left(-\frac{a\rho}{RT}\right) \cdot \left(-\frac{-a\rho}{RT^2}\right) \Rightarrow \left(\frac{\partial Z}{\partial T}\right)_\rho = P(y) \exp\left(-\frac{a\rho}{RT}\right) \frac{a\rho}{RT^2}$$

Now we Can Put all of that in the formula for enthalpy:

$$\frac{H^R}{RT} = (Z - 1) - T \int_0^\rho \frac{a\rho}{RT^2} Z \frac{d\rho}{\rho}$$

we Can Cancel out the density and Change everything to be in terms of y

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho Z d\rho$$

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho Z(y) d\rho$$

$$\boxed{\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a\rho}{RT}\right) d\rho \quad y = \frac{b\rho}{4}}$$

And to get the total molar enthalpy:

$$H = H^{IG} + H^R$$

where H^{IG} we Can get using the CHE Calculator and we Can just solve for H^R from the formula we just derived.

6.3 Molar Entropy, S^R

In Chapter 4 we are given the formula for the residual entropy as follows:

$$\boxed{\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi}$$

which is just the previous formulas we derived

$$\begin{aligned}\frac{S^R}{R} &= (Z - 1) - \frac{a}{RT} \int_0^\rho \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a\rho}{RT}\right) d\rho \\ &\quad - (Z - 1) - \ln Z + \int_0^\rho \left[\frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a\rho}{RT}\right) - 1 \right] \frac{d\rho}{\rho} \\ \text{and} \quad y &= \frac{b\rho}{4}\end{aligned}$$

And to get the total molar entropy:

$$S = S^{IG} + S^R$$

where S^{IG} we Can get using the CHE Calculator and we Can just solve for S^R from the formula we just derived.

7 Finding β , C_p , C_v , μ_{JT} , and v

7.1 Thermal Expansion Coefficient, β , K^{-1}

In Chapter 4 we are given the following formula for β :

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P = \frac{1}{\rho} \frac{\left(\frac{\partial P}{\partial T} \right)_\rho}{\left(\frac{\partial P}{\partial \rho} \right)_T} = \frac{1}{T} \frac{C_1}{C_2}$$

From the book we know that:

$$\left(\frac{\partial P}{\partial T} \right)_\rho = \rho R \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_\rho \right] \quad \text{we previously has solved for } \left(\frac{\partial Z}{\partial T} \right)_\rho$$

and we got

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{Z\rho q}{T} \quad \text{where } Z = P(y) \exp(-q\rho) \quad \text{where}$$

$$P(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}, \quad q = \frac{a}{RT}, \quad \text{and } \rho = \frac{4y}{b} \quad \text{and therefore we get}$$

$$\left(\frac{\partial P}{\partial T} \right)_\rho = \frac{4yR}{b} \left[Z + Z \frac{4aqy}{RTb} \right] = \rho R Z (1 + q\rho)$$

In the book we also know:

$$\left(\frac{\partial P}{\partial \rho} \right)_T = RT \left(Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T \right)$$

to Solve for $\left(\frac{\partial Z}{\partial \rho}\right)_T$ we will do the following:

$$Z = P(y) \exp(-q\rho)$$

and since Z is written in terms of y and y depends on ρ we will do the Chain rule:

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = \frac{\partial}{\partial \rho} [P(y)e^{-q\rho}] \Rightarrow e^{-q\rho} \frac{dP}{dy} \frac{dy}{d\rho} + P(y) \frac{d}{d\rho} e^{-q\rho}$$

$$\Rightarrow \frac{\partial y}{\partial \rho} = \frac{b}{4} \Rightarrow \left(\frac{\partial y}{\partial \rho}\right)_T = \frac{b}{4}$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = e^{-q\rho} P'(y) \frac{b}{4} - qP(y)e^{-q\rho}$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = e^{-q\rho} \left[P'(y) \frac{b}{4} - qP(y) \right], \quad Z = P(y)e^{-q\rho}$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = Z \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q \right]$$

No we will Plug everything in $\left(\frac{\partial P}{\partial \rho}\right)_T$

$$\left(\frac{\partial P}{\partial \rho}\right)_T = RT \left(Z + \rho Z \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q \right] \right) \quad \text{we know that } Z = P(y) \exp(-q\rho) \text{ so}$$

$$\left(\frac{\partial P}{\partial \rho}\right)_T = RT \left(Z + \rho Z \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q \right] \right) = RTZ \left(1 + \rho \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q \right] \right)$$

So now we Can Solve for β ,

$$\beta = \frac{1}{\rho} \left[\frac{\rho RZ(1+q\rho)}{RTZ \left(1 + \rho \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q \right] \right)} \right] \Rightarrow \beta = \frac{1}{T} \frac{(1+q\rho)}{1 + \rho \left(\frac{P'(y)}{P(y)} \frac{b}{4} - q \right)}$$

$$\boxed{\beta = \frac{1}{T} \frac{(1+q\rho)}{1 + \rho \left(\frac{P'(y)}{P(y)} \frac{b}{4} - q \right)} \quad \text{where } \frac{P'(y)}{P(y)} = \frac{1+2y-3y^2}{1+y+y^2-y^3} + \frac{3}{1-y}}$$

7.2 Molar Heat Capacity, C_v , $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

In the book we are given

$$C_v = C_v^{IG} + C_v^R$$

and we Can ge C_v^{IG} from the CHE Calculator and

$$C_v^R = \int_{V=\infty}^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV$$

we Can Change V to ρ by doing the following

$$V = \frac{1}{\rho} \quad dV = -\frac{d\rho}{\rho^2} \quad \text{and the integral becomes}$$

$$C_v^R = \int_0^\rho T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho \left(-\frac{d\rho}{\rho^2} \right) \quad \text{and we are given that} \quad \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \rho R \left[2 \left(\frac{\partial Z}{\partial T} \right)_\rho + T \left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho \right]$$

we already found

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{4aqyZ}{RT^2b} \quad \text{we also know } \rho = \frac{4y}{b}$$

So

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{a\rho}{RT^2} Z$$

to find $\left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho$

$$\left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho = \frac{\partial}{\partial T} \left(\frac{a\rho}{RT^2} \right) Z + \frac{a\rho}{RT^2} \left(\frac{\partial Z}{\partial T} \right)_\rho \Rightarrow -\frac{2a\rho}{RT^3} Z + \frac{a\rho}{RT^2} \left(\frac{a\rho}{RT^2} \right) Z$$

then

$$\left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho = Z \left[-\frac{2a\rho}{RT^3} + \frac{a^2\rho^2}{R^2T^4} \right]$$

So now we Can Plug Everything in:

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \rho R \left[2 \frac{a\rho}{RT^2} Z + T Z \left(-\frac{2a\rho}{RT^3} + \frac{a^2\rho^2}{R^2T^4} \right) \right]$$

To Simplify things we know $\rho = \frac{4y}{b}$ and $q = \frac{a}{RT}$ so,

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \rho R \left[2 \left(\frac{a\rho}{RT^2} \right) Z + T Z \left(-2 \left(\frac{a\rho}{RT^3} \right) + \frac{a^2\rho^2}{R^2T^4} \right) \right]$$

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \rho R \left[2 \frac{q\rho}{T} Z + T Z \left(-2 \frac{q\rho}{T^2} + \frac{q^2\rho^2}{T^2} \right) \right]$$

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \frac{\rho R Z}{T} [2q\rho - 2q\rho + q^2\rho^2]$$

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_\rho = \frac{\rho R Z q^2 \rho^2}{T}$$

now we Can Plug Everything into the integral:

$$C_v^R = - \int_0^\rho T \frac{\rho R Z q^2 \rho^2}{T} \frac{d\rho}{\rho^2} \Rightarrow C_v^R = -Rq^2 \int_0^\rho Z \rho d\rho \Rightarrow C_v^R = -Rq^2 \int_0^\rho Z \rho d\rho$$

$$\boxed{C_v^R = -Rq^2 \int_0^\rho \frac{1+y+y^2-y^3}{(1-y)^3} \exp\left(\frac{a\rho}{RT}\right) \rho d\rho \quad \text{and } y = \frac{b\rho}{4}}$$

7.3 Molar Heat Capacity, C_p , $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

First we need to find

$$C_p - C_v = \frac{T}{\rho^2} \frac{\left[\left(\frac{\partial P}{\partial T}\right)_\rho\right]^2}{\left(\frac{\partial P}{\partial \rho}\right)_T} = R \frac{C_1^2}{C_2}$$

where

$$C_1 = Z + T \left(\frac{\partial Z}{\partial T}\right)_\rho, \quad \left(\frac{\partial Z}{\partial T}\right)_\rho = \frac{a\rho}{RT^2} Z = \frac{Zq\rho}{T} \Rightarrow C_1 = Z + Zq\rho$$

$$C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho}\right)_T, \quad \left(\frac{\partial Z}{\partial \rho}\right)_T = Z \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q\right] \Rightarrow C_2 = Z + Z \left[y \frac{P'(y)}{P(y)} - q\rho\right]$$

So,

$$C_p - C_v = R \left[\frac{(Z + Zq\rho)^2}{Z + Z \left(y \frac{P'(y)}{P(y)} - q\rho\right)} \right] \Rightarrow C_p - C_v = R \left[\frac{Z(1 + q^2\rho^2)}{1 + y \left(\frac{P'(y)}{P(y)}\right) - q\rho} \right]$$

Now Solve for C_p

$$\boxed{C_p = C_v + R \left[\frac{Z(1 + q^2\rho^2)}{1 + y \left(\frac{P'(y)}{P(y)}\right) - q\rho} \right] \quad \text{where } \frac{P'(y)}{P(y)} = \frac{1 + 2y - 3y^2}{1 + y + y^2 - y^3} + \frac{3}{1 - y}}$$

7.4 Joule Thomson Coefficient, μ_{JT} , K/bar

$$\mu_{JT} = \frac{\beta T - 1}{\rho C_p} \quad \text{now we have to write it in terms of } y \text{ using } y = \frac{b\rho}{4} \Rightarrow \rho = \frac{4y}{b}$$

$$\boxed{\mu_{JT} = \frac{(\beta T - 1)}{\rho C_p}}$$

7.5 Speed of Sound, v , m/s

$$v^2 = \frac{1}{M_w} \frac{C_p}{C_v} \left(\frac{\partial P}{\partial \rho} \right)_T \quad \text{we have previously found } \left(\frac{\partial P}{\partial \rho} \right)_T$$

$$\left(\frac{\partial P}{\partial \rho} \right)_T = RTZ \left(1 + y \left[\frac{P'(y)}{P(y)} - q \right] \right)$$

$$\boxed{v = \sqrt{\frac{1}{M_w} \frac{C_p}{C_v} RTZ \left(1 + y \left[\frac{P'(y)}{P(y)} - q \right] \right)}}$$

8 finding Secondary Properties using multiple methods

8.1 Using Experimental Data Calculated in Assignment #3

$$H = -2.125 \frac{\text{kJ}}{\text{mol}}$$

$$S = -75.559 \frac{\text{kJ}}{\text{kmol K}}$$

$$Z = 1.786$$

$$\rho = 19.723 \frac{\text{kmol}}{\text{m}^3}$$

$$\beta = 0.00167 \text{ K}^{-1}$$

$$C_v = 23.325 \frac{\text{kJ}}{\text{kmol K}}$$

$$C_p = 33.207 \frac{\text{kJ}}{\text{kmol K}}$$

$$\mu_{JT} = -0.0804 \frac{\text{K}}{\text{bar}}$$

$$v = 796.889 \frac{\text{m}}{\text{s}}$$

we also were able to find

$$T_r = \frac{T}{T_c} = \frac{283.15 \text{ K}}{126.20 \text{ K}} = 2.243660856 \quad \text{and} \quad P_r = \frac{P}{P_c} = \frac{830 \text{ bar}}{33.94 \text{ bar}} = 24.452$$

8.2 Using the Peng-Robinson EOS in the CHE Calculator

To Calculate A and B we know from the book that

$$A = \lambda_1 \sigma(T_r) \frac{P_r}{T_r^2}$$

$$B = \lambda_2 \frac{P_r}{T_r}$$

$$\sigma(T_r) = \left[1 + \kappa(\omega) \left(1 - \sqrt{T_r} \right) \right]^2$$

$$\kappa(\omega) = 0.37464 + 1.54226\omega - 0.26992\omega^2$$

where $\omega = 0.0372$ which is the Acentric factor given in NIST

$$\lambda_1 = 0.457235$$

$$\lambda_2 = 0.077796$$

$$Z_c = 0.3074$$

we can Calculate everything to get A and B in the CHE Calculator but first we would need to solve for T_r in terms of Kelvin So

$$T_r = \frac{283.15 \text{ K}}{126.2 \text{ K}} = 2.2437$$

$$\kappa(\omega) = 0.43164, \quad \sigma = 0.61637, \quad A = 1.36894, \quad B = 0.04785$$

Now we can use the functions in the CHE Calculator to find:

$$\text{Compressibility Factor} \Rightarrow zComp(N_2, 10^\circ C, 830\text{bar}, 1) = Z = 1.62178$$

Since a liquid Phase will give an error

$$\text{fugacity Coefficient} \Rightarrow \Phi_c(N_2, N_2, 10^\circ C, 830\text{bar}, 2) = \phi = 1.31787$$

$$\text{Molar Enthalpy} \Rightarrow \frac{\text{Enthalpy}(N_2, 10^\circ C, 830\text{bar}, 2)}{1000} = H = -1.573177 \frac{\text{kJ}}{\text{mol}}$$

$$\text{Molar Entropy} \Rightarrow \text{Entropy}(N_2, 10^\circ C, 830\text{bar}, 2) = S = -63.58694 \frac{\text{kJ}}{\text{kmol K}}$$

Now to Calculate C_p , C_v , μ_{JT} , and v we will just use the function PProp to give us

$$\beta = 0.002459 \text{ K}^{-1}$$

$$C_v = 24.70498 \frac{\text{kJ}}{\text{kmol K}}$$

$$C_p = 39.1225 \frac{\text{kJ}}{\text{kmol K}}$$

$$\mu_{JT} = -0.03572 \frac{\text{K}}{\text{bar}}$$

$$v = 689.9679 \frac{\text{m}}{\text{s}}$$

For Peng-Robinson	
Λ_1	0.457235
Λ_2	0.077796
w	0.0372
k(w)	0.431638546
σ	0.616371715
Cp	39.12253383
Cv	24.7049835
Speed of Sound	689.9679457
Joule Thomson Coefficient	-0.035715119
Thermal Expansion Coefficient	0.002458951
	3.827524774

Figure 5: Calculations Needed for Peng Robinson

8.3 Using the Exponential EOS with Theoretical Values of λ_1 and λ_2

$$\lambda_1 = 0.8144 \quad \text{and} \quad \lambda_2 = 0.38907$$

$$a = \lambda_1 \frac{(RT_c)^2}{P_c} \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Use the CHE Calculator to solve:

$$a = 0.8144 \frac{\left[\left(8.314 \frac{\text{kJ}}{\text{kmol K}} \right) (126.2 \text{ K}) \right]^2}{33.94 \text{ bar}} \left(\frac{1 \text{ bar} \cdot \text{m}^3}{100 \text{ kJ}} \right) = 264.137 \frac{\text{kJ} \cdot \text{m}^3}{\text{kmol}^2}$$

$$b = 0.38907 \frac{\left(8.314 \frac{\text{kJ}}{\text{kmol K}} \right) (126.2 \text{ K})}{33.94 \text{ bar}} \left(\frac{1 \text{ bar} \cdot \text{m}^3}{100 \text{ kJ}} \right) = 0.12026 \frac{\text{m}^3}{\text{kmol}}$$

we have Previously Calculated

$$Z_c = 0.2545$$

Now to find Z we will do the following:

start w/ a Z guess eg.

$$Z = 1$$

find a $V \Rightarrow$

$$V = \frac{ZRT}{P} \left(\frac{1 \text{ bar} \cdot \text{m}^3}{100 \text{ kJ}} \right) \Rightarrow V = 0.02836 \text{ m}^3$$

find a $Y \Rightarrow$

$$Y = \frac{b}{4V} \Rightarrow Y = 1.0600$$

find a $Z_{calc} \Rightarrow$

$$Z = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp \left(-\frac{a}{VRT} \right) \Rightarrow Z_{calc} = -176.064$$

Do an error

$$Z_{calc} - Z_{guessed} = -177.064$$

Now goal seek the error by Changing guessed Z to give us

$$Z = 2.077$$

Now we can find density:

$$\rho = \frac{P}{ZRT} \left(\frac{100 \text{ kJ}}{1 \text{ bar} \cdot \text{m}^3} \right) \Rightarrow \rho = 16.9738 \frac{\text{kmol}}{\text{m}^3}$$

Now let's find derivatives:

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = \frac{qyz}{T}$$

$$\rho \left(\frac{\partial Z}{\partial \rho}\right)_T = y \left[\frac{P'(y)}{P(y)} - q \right] Z$$

$$\left(\frac{\partial P}{\partial T}\right)_\rho = \frac{4yR}{b}Z(1+qy), \quad \left(\frac{\partial P}{\partial T}\right)_\rho = C_1, \quad \left(\frac{\partial P}{\partial T}\right)_\rho = \rho R \left[Z + T \left(\frac{\partial Z}{\partial T}\right)_\rho \right]$$

$$\left(\frac{\partial P}{\partial \rho}\right)_T = RTZ \left(1 + y \left[\frac{P'(y)}{P(y)} - q \right] \right), \quad \left(\frac{\partial P}{\partial \rho}\right)_T = C_2, \quad \left(\frac{\partial P}{\partial \rho}\right)_T = RT \left[Z + \rho \left(\frac{\partial Z}{\partial \rho}\right)_T \right]$$

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_\rho = -\frac{\rho RZ}{T}q^2y^2, \quad \left(\frac{\partial^2 P}{\partial T^2}\right)_\rho = -\rho R \left[2 \left(\frac{\partial Z}{\partial T}\right)_\rho + T \left(\frac{\partial^2 Z}{\partial T^2}\right)_\rho \right]$$

$$\left(\frac{\partial^2 Z}{\partial T^2}\right)_\rho = Z \left[-\frac{2qy}{T^2} + \frac{q^2y^2}{T^2} \right]$$

variables to keep in mind:

$$P(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}$$

$$q = \frac{4a}{RTb}$$

$$\rho = \frac{4y}{b}$$

Now that we have all those derivatives in terms of known values we Can Calculate them in the Calculator. once we have that we Can start Solving for unknowns.

8.3.1 Fugacity Coefficient, ϕ

we have the following formula for fugacity:

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \left[\frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a\rho}{RT}\right) - 1 \right] \frac{d\rho}{\rho} \quad \text{and} \quad y = \frac{b\rho}{4}$$

we have previously found $Z = 2.077$ and $\rho = 16.9738$ we now need to expand the integrand in terms of ρ instead of y and we get:

$$\int_0^\rho \left\{ \left[\frac{1 + \frac{b\rho}{4} + \frac{b^2\rho^2}{16} - \frac{b^3\rho^3}{64}}{\left(1 - \frac{b\rho}{4}\right)^3} \right] \exp\left(-\frac{a\rho}{RT}\right) - 1 \right\} \frac{d\rho}{\rho}$$

where

$$\frac{a}{RT} = q = 0.1122 \quad \text{and} \quad b = 0.12026$$

$$\frac{b}{4} = k = 0.030066$$

$$\Rightarrow \int_0^\rho \left[\frac{1 + k\rho + k^2\rho^2 - k^3\rho^3}{(1 - k\rho)^3} \exp(-q\rho) - 1 \right] \frac{d\rho}{\rho}$$

well past this into Wolfram Alpha to Compute the integral we make sure to plug all the numbers in to give us an integral value of

$$0.486917$$

Now to find the fugacity Coefficient:

$$\phi = \exp((2.077 - 1) - \ln(2.077) + 0.486917) \Rightarrow \phi = 2.30039$$

8.3.2 Enthalpy, H , $\frac{\text{kJ}}{\text{mol}}$

We know

$$H = H^R + H^{IG}$$

we Can get H^{IG} from the CHE Calculator to give us:

$$H^{IG} = -437.7630 \frac{\text{kJ}}{\text{kmol}}$$

we know that

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a\rho}{RT}\right) d\rho \quad y = \frac{b\rho}{4}$$

we Can expand the integral in terms of ρ to give us:

$$\frac{a}{RT} \int_0^\rho \frac{1 + \frac{b\rho}{4} + \frac{b^2\rho^2}{16} - \frac{b^3\rho^3}{64}}{\left(1 - \frac{b\rho}{4}\right)^3} \exp\left(-\frac{a\rho}{RT}\right) d\rho \Rightarrow q \int_0^\rho \frac{1 + k\rho + k^2\rho^2 - k^3\rho^3}{(1 - k\rho)^3} \exp(-q\rho) d\rho$$

we Can put it in Integral Calculator to give us an integral value of

$$2.5044382$$

No we plug everything in

$$H^R = RT(2 - 1 - 2.5044382) \Rightarrow H^R = -3359.954179 \frac{\text{kJ}}{\text{kmol}}$$

So,

$$H = H^R + H^{IG} \Rightarrow H = -3.7977 \frac{\text{kJ}}{\text{mol}}$$

8.3.3 Entropy, S , $\frac{\text{kJ}}{\text{kmol K}}$

we Know

$$S = S^R + S^{IG}$$

we Can get S^{IG} from the CHE Calculator to give us:

$$S^{IG} = -57.28 \frac{\text{kJ}}{\text{kmol K}}$$

we know that

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi \Rightarrow S^R = \frac{H^R}{T} - R \ln \phi$$

we have previously Calculated everything so we can plug them in:

$$\frac{H^R}{T} = -0.0134 \quad \text{and} \quad R \ln \phi = 19.125$$

So,

$$S = S^R + S^{IG} \Rightarrow S = -76.4209 \frac{\text{kJ}}{\text{kmol K}}$$

8.3.4 Heat Capacity at Constant Volume, C_v , $\frac{\text{kJ}}{\text{kmol K}}$

$$C_v = C_v^{IG} + C_v^R$$

from the CHE Calculator

$$C_v^{IG} = 20.877 \frac{\text{kJ}}{\text{kmol K}}$$

And we know that:

$$C_v^R = -Rq^2 \int_0^\rho \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp(-q\rho) \cdot \rho d\rho, \quad y = \frac{b\rho}{4}$$

we can expand the integral in terms of ρ to give us:

$$C_v^R = -Rq^2 \int_0^\rho \frac{1 + \frac{b\rho}{4} + \frac{b^2\rho^2}{16} - \frac{b^3\rho^3}{64}}{\left(1 - \frac{b\rho}{4}\right)^3} \exp(-q\rho) \cdot \rho d\rho \Rightarrow C_v^R = -Rq^2 \int_0^\rho \frac{1 + k\rho + k^2\rho^2 - k^3\rho^3}{(1 - k\rho)^3} \exp(-q\rho) \cdot \rho d\rho$$

we can put it in Integral Calculator to give us an integral value of

$$212.4932$$

Now we plug everything in

$$C_v^R = -Rq^2(212.4932) \Rightarrow C_v^R = -22.241$$

So,

$$C_v = C_v^R + C_v^{IG} \Rightarrow C_v = -1.38 \frac{\text{kJ}}{\text{kmol K}}$$

Since this method gives us a negative C_v we will try another method keeping the original C_v^R integral

$$\begin{aligned} C_v^R &= \int_{\infty}^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV \\ \left(\frac{\partial^2 P}{\partial T^2} \right)_V &\Rightarrow P = \frac{RTZ}{V} \Rightarrow \frac{\partial P}{\partial T} = \frac{R}{V} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_V \right] \\ \left(\frac{\partial^2 P}{\partial T^2} \right)_V &= \frac{R}{V} \left[\left(\frac{\partial Z}{\partial T} \right)_V + \frac{\partial}{\partial T} \left(T \left(\frac{\partial Z}{\partial T} \right)_V \right) \right] \Rightarrow \frac{R}{V} \left[2 \left(\frac{\partial Z}{\partial T} \right)_V + T \left(\frac{\partial^2 Z}{\partial T^2} \right)_V \right] \end{aligned}$$

Now to find:

$$\left(\frac{\partial Z}{\partial T} \right)_V$$

$$Z = P(y) \exp \left(-\frac{a}{RTV} \right), \quad y = \frac{b}{4V}, \quad \text{and we will let } Q = \frac{a}{RTV}$$

and the exponential is the only term that depends on T So,

$$\left(\frac{\partial Z}{\partial T} \right)_V = \frac{Za}{RT^2V} = \frac{ZQ}{T}$$

So,

$$\begin{aligned} \left(\frac{\partial^2 Z}{\partial T^2} \right)_V &= Z \cdot \frac{a}{RT^2V} \cdot \frac{a}{RT^2V} - \frac{2Za}{RT^3V} = \frac{ZQ^2}{T^2} - \frac{2ZQ}{T^2} \\ \left(\frac{\partial^2 P}{\partial T^2} \right)_V &= \frac{R}{V} \left[2 \frac{ZQ}{T} + T \left(\frac{ZQ^2}{T^2} - \frac{2ZQ}{T^2} \right) \right] \Rightarrow \frac{R}{V} \left[\frac{ZQ^2}{T} \right] \end{aligned}$$

So,

$$C_v^R = \int_{\infty}^V T \frac{R}{V} \left[\frac{ZQ^2}{T} \right] dV \Rightarrow \int_{\infty}^V \frac{RZ}{V} \frac{a^2}{R^2 T^2 V^2} dV \Rightarrow \int_{\infty}^V \frac{Za^2}{RT^2 V^3} dV$$

Let's write Z in terms of V Since

$$Z = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp \left(-\frac{a}{RTV} \right) \quad \text{and} \quad y = \frac{b}{4V}$$

and

$$k = \frac{b}{4} \quad \text{and} \quad q = \frac{a}{RT}$$

So,

$$C_v^R = \int_{\infty}^{V_f} \frac{1 + \frac{k}{V} + \frac{k^2}{V^2} - \frac{k^3}{V^3}}{\left(1 - \frac{k}{V}\right)^3} \exp\left(-\frac{q}{V}\right) \cdot \frac{a^2}{RT^2 V^3} dV$$

we know

$$V_f = 0.058914164 \text{ m}^3$$

So we can Plug this integral into Wolfram Alpha to give us a value of

$$-22.2413$$

So

$$C_v = C_v^{IG} + C_v^R \Rightarrow C_v = -1.36 \frac{\text{kJ}}{\text{kmol K}}$$

So either way this integral gives us a wrong C_v so therefore we can't use it for any of the Calculations so we will use the C_v obtained from the NIST Data which is

$$C_v = 23.821 \frac{\text{kJ}}{\text{kmol K}}$$

8.3.5 Heat Capacity at Constant Volume, C_p , $\frac{\text{kJ}}{\text{kmol K}}$ Using C_v from NIST

we know that:

$$C_p = C_v + R \frac{C_1^2}{C_2}$$

$$C_1 = Z + T \left(\frac{\partial Z}{\partial T} \right)_{\rho}, \quad \frac{\partial Z}{\partial T} = \frac{zq\rho}{T} \Rightarrow C_1 = Z + Zq\rho$$

$$C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T, \quad \frac{\partial Z}{\partial \rho} = Z \left[\frac{P'(y)}{P(y)} \frac{b}{4} - q \right] \Rightarrow C_2 = Z + Z \left[y \frac{P'(y)}{P(y)} - q\rho \right]$$

and we have Calculated everything in the CHE Calculator

$$C_1 = 6.033$$

$$C_2 = 5.418$$

So,

$$C_p = C_v + R \frac{C_1^2}{C_2} \Rightarrow C_p = 79.676 \frac{\text{kJ}}{\text{kmol K}}$$

8.3.6 Thermal Expansion Coefficient, β , K^{-1}

$$\beta = \frac{1}{T} \frac{(1 + q\rho)}{1 + \rho \left(\frac{P'(y)}{P(y)} - q \right)} \quad \text{where} \quad \frac{P'(y)}{P(y)} = \frac{1 + 2y - 3y^2}{1 + y + y^2 - y^3} + \frac{3}{1 - y}$$

or

$$\beta = \frac{1}{T} \frac{C_1}{C_2}$$

Since everything is already Calculated in the Calculator we can just Plug everything in to get

$$\beta = 0.00393 \text{ K}^{-1}$$

8.3.7 Joule Thomson Coefficient, μ_{JT} , $\frac{\text{K}}{\text{bar}}$ Using C_v from NIST

$$\mu_{JT} = \frac{(\beta T - 1)}{\rho C_p} \times \frac{100 \text{ kJ}}{1 \text{ m}^3 \cdot \text{bar}}$$

Plug everything to the CHE Calculator we get:

$$\mu_{JT} = 0.008396 \frac{\text{K}}{\text{bar}}$$

8.3.8 Speed of Sound, v , m/s Using C_v from NIST

$$v = \sqrt{\frac{1}{M_w} \frac{C_p}{C_v} R T Z \left(1 + y \left[\frac{P'(y)}{P(y)} - q \right] \right)}$$

or

$$v = \sqrt{\frac{R T}{M_w} \frac{C_p}{C_v} C_2 \cdot 1000}$$

plug everything into the CHE Calculator to get

$$v = 1234.05592 \frac{\text{m}}{\text{s}}$$

For Theo. Values	
Λ_1	0.814426965
Λ_2	0.389071619
R, KJ/Kmol K	8.314
Tc	126.2
Pc	33.943875
KJ to m ³ *bar	100
$g, 4a/RTb$	3.731860259
$q, a/RT$	0.112202592
$k, b/4$	0.030066129
V	0.058914164
Y	0.510337871
Zcalc	2.077166343
error	0.00
Fugacity Integral	0.486917
Enthalpy:	
HIG	-437.7625975
Integral Value	2.50444
HR	-3359.958496
Entropy:	
HR/T	-0.0134124
$R \cdot \ln \phi$	19.12544072
SR	-19.13885312
SIG	-57.28208751
Cv	
CvIG	20.87745872
CR Integral	212.493
CR	-22.24131321
$P'(y)/P(y)$	6.883353347

Figure 6: Calculations Needed for the Theoretical Values

8.4 Using the exponential EOS with Experimental values of λ_1 and λ_2

$$\lambda_1 = 0.26329 \quad \lambda_2 = 0.176634$$

since we have all the cells referenced correctly we can just drag the cells to get the following results:

$$a = 85.3918$$

$$b = 0.054599$$

$$Z = 1.7085$$

$$\phi = 2.00323$$

$$H = -1.039 \frac{\text{kJ}}{\text{mol}}$$

$$S = -73.9407 \frac{\text{kJ}}{\text{kmol K}}$$

$$C_1 = 2.98745$$

$$C_2 = 2.91599$$

$$\rho = 20.636$$

$$\beta = 0.003618 \text{ K}^{-1}$$

$$C_v = 17.609 \frac{\text{kJ}}{\text{kmol K}}$$

$$C_p = 43.055 \frac{\text{kJ}}{\text{kmol K}}$$

$$\mu_{JT} = 0.00278 \frac{\text{K}}{\text{bar}}$$

$$v = 774.058 \text{ m/s}$$

The experimental values came out to be more accurate then the theoretical ones Since they were fit to the actual data at that state.

	For Exp. Values
Λ_1	0.263292745
Λ_2	0.176634575
R, KJ/Kmol K	8.314
Tc	126.2
Pc	33.943875
KJ to m ³ *bar	100
$g, 4a/RTb$	2.657455228
$q, a/RT$	0.036273515
$k, b/4$	0.013649718
V	0.048458884
Y	0.281676275
Zcalc	1.70853914
error	0.00
Fugacity Integral	0.521864
Enthalpy:	
HIG	-437.7625975
Integral Value	0.964053
HR	-601.5076386
Entropy:	
HR/T	-0.003670388
$R \cdot \ln \phi$	16.65491313
SR	-16.65858352
SIG	-57.28208751
Cv	
CvIG	20.87745872
CR Integral	298.788
CR	-3.268529779
P'(y)/P(y)	5.166423958

Figure 7: Calculations Needed for the Experimental Values

9 Conclusion

This report used several approaches to calculate the thermodynamic properties of nitrogen and to compare their results. The theoretical values of λ_1 and λ_2 were obtained from the critical thermodynamic constraints of the exponential EOS, while the experimental values were obtained by fitting the equation of state to data. The fitted values produced better agreement with the property data at the specified state, but they did not satisfy the critical constraints of the model as well as the theoretical values.

The report also showed how the fugacity coefficient, enthalpy, entropy, and secondary properties can be derived in terms of density and then evaluated numerically. A comparison with the Peng-Robinson EOS provided an additional point of reference for the results. One important observation was that the direct EOS-based calculation of C_v produced a non-physical negative value, which indicated a limitation in that part of the method. Because of this, the NIST value of C_v was used in the later property calculations where needed.

Overall, this assignment demonstrated the connection between thermodynamic theory, equations of state, numerical fitting, and property estimation. It also showed that different methods may perform differently depending on whether the goal is mathematical consistency at the critical point or better agreement with data at a specific condition.

Summary Tables

Model Parameters and Compressibility Summary

Table 1: Summary of EOS parameters and compressibility values

Quantity	Experimental Data	Peng-Robinson EOS	Dieterici EOS Theoretical Constants	Dieterici EOS Experimental Constants
λ_1	—	—	0.81442	0.26329
λ_2	—	—	0.38907	0.17663
Critical Compressibility Factor, Z_c	—	0.3074	0.254540144	—
Compressibility Factor, Z	1.787590612	1.621784282	2.077166092	1.708539082

Secondary Properties Summary

Table 2: Summary of secondary properties at $T = 10^\circ\text{C}$ and $P = 830$ bar

Property	Units	Experimental Data	Peng-Robinson EOS	Dieterici EOS Theoretical Constants	Dieterici EOS Experimental Constants	NIST Value
Fugacity Coefficient, ϕ	—	—	1.317868366	2.300389791	2.003237085	—
Molar Enthalpy, H	kJ/mol	-2.124659675	-1.573177117	-3.797721094	-1.039270236	10.884
Molar Entropy, S	kJ/(kmol · K)	-75.55888391	-63.58694857	-76.42094063	-73.94067102	48.523
Density, ρ	kmol/m ³	19.72347533	21.73873429	16.97384695	20.63605081	19.596
Thermal Expansion Coefficient, β	1/K	0.001671236	0.002458951	0.003932732	0.003618245	—
Heat Capacity at Constant Volume, C_v	kJ/(kmol · K)	23.32511018	24.7049835	-1.363854491	17.60892894	23.821
Heat Capacity at Constant Pressure, C_p	kJ/(kmol · K)	33.20702547	39.12253383	79.67637438	43.05528584	38.731
Joule-Thomson Coefficient, μ_{JT}	K/bar	-0.080431055	-0.035715119	0.008396318	0.00275818	-0.047978
Speed of Sound, v	m/s	796.8890813	689.9679457	1234.05592	774.0579263	778.5

*For the exponential EOS with theoretical values of $\{\lambda_1, \lambda_2\}$, the report used the NIST value of C_v because the direct EOS calculation gave a non-physical negative result.

Property	Units	Value from Experimental Data	Peng-Robinson EOS	Dieterici EOS Theoretical Constants	Dieterici EOS Experimental Constants	NIST Value
Temperature	C	10.00	10.00	10.00	10.00	10.00
Pressure	bar	830.00	830.00	830.00	830.00	830.00
Temperature	K	283.15	283.15	283.15	283.15	283.15
Reduced Temperature		2.243660856	2.243660856	2.243660856	2.243660856	2.243660856
Reduced Pressure		24.4521287	24.4521287	24.4521287	24.4521287	24.4521287
A			1.368940016	264.1371424	85.39181086	
B			0.84784552	0.120264516	0.054598872	
Critical Compressibility Factor			0.3074	0.254540144		
Compressibility Factor		1.787590612	1.621784282	2.077166092	1.708539082	
Equation of State			Peng-Robinson			
Fugacity Coefficient			1.317868366	2.300389791	2.003237085	
Molar Enthalpy	kJ/mol	-2.124659675	-1.573177117	-3.797721094	-1.039270236	10.884
Molar Entropy	kJ/(kmol K)	-75.55888391	-63.58694857	-76.42094063	-73.94067102	48.523
T dZ/dT at constant ρ				0.013971333	0.004516735	
ρ dZ/d ρ at constant T				0.196818129	0.058511852	
$C_v = Z + T$ dZ/dT				6.033148897	2.9874525	
$C_p = Z + \rho$ dZ/d ρ				5.417926892	2.915992641	
Residual C_v/R						
Density	kmol/m ³	19.72347533	21.73873429	16.97384695	20.63605081	19.596
Thermal Expansion Coefficient	1/K	0.001671236	0.002458951	0.003932732	0.003618245	
Heat Capacity at Constant Volume	kJ/(kmol K)	23.32511018	24.7049835	-1.363854491	17.60892894	23.821
Heat Capacity at Constant Pressure	kJ/(kmol K)	33.20702547	39.12253383	79.67637438	43.05528584	38.731
Joule-Thomson Coefficient	K/bar	-0.080431055	-0.035715119	0.008396318	0.00275818	-0.047978
Speed of Sound	m/s	796.8890813	689.9679457	1234.05592	774.0579263	778.5

Figure 8: Final Excel Table

References

- [1] Michael Caracotsios, *An Introduction to Chemical Engineering Thermodynamics: Theory and Applications*, First Edition, Chapter 3: “Thermal Properties of Pure Fluids,” University of Illinois Chicago, Spring 2026.
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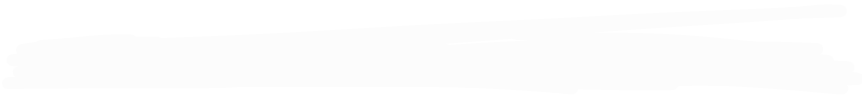
Assignment #03

CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS

Spring 2026

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Abstract

This report presents solutions for CHE 301 Assignment #03, consisting of Problem A and Problem B. In Problem A, thermodynamic properties were evaluated at 10°C and 830 bar using density-based relationships and Akima interpolation (FAKIMA) to obtain needed derivatives and integrands. Properties computed include β , ζ , enthalpy and entropy changes by integration, C_p and C_v , the Joule–Thomson coefficient, and the speed of sound. In Problem B, ideal-gas venting through a narrow pipe section was analyzed under reversible adiabatic (isentropic) assumptions to determine tank mass, critical-section temperature and pressure, pipe diameter, and masses during venting.

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1 Introduction

This assignment applies core chemical engineering thermodynamics tools to compute real-gas properties and to analyze compressible flow under ideal-gas/isentropic assumptions. Problem A focuses on evaluating properties at a specified state (10°C and 830 bar) using tabulated data, interpolation, and thermodynamic identities written in terms of density. Problem B analyzes the venting of ethylene from a tank through a narrow pipe section, including the critical (sonic) condition and associated flow/geometry requirements.

The objective of this report is to present the calculations, intermediate steps, and final results in a clear, typed format while preserving the original workflow used to obtain the reported numerical values.

2 Background

Thermodynamic properties for real fluids can be obtained from tabulated data and evaluated at intermediate states using interpolation. In this report, Akima interpolation (FAKIMA) is used to estimate values and derivatives needed for property calculations.

For Problem A, the analysis uses density-based identities and partial-derivative relations to compute:

- The compressibility factor Z at the target state using interpolation and averaging approaches.
- The thermal expansion coefficient β from $\left(\frac{\partial \rho}{\partial T}\right)_P$.
- The coefficient ζ from $\left(\frac{\partial P}{\partial T}\right)_\rho$ using derivative transformations.
- Enthalpy and entropy changes using integrals written in terms of pressure (and later density) with $V = \frac{1}{\rho}$.
- Heat capacities C_p and C_v using thermodynamic identities involving $\left(\frac{\partial P}{\partial T}\right)_\rho$ and $\left(\frac{\partial P}{\partial \rho}\right)_T$, plus a density-form integral for C_v requiring second derivatives.
- The Joule–Thomson coefficient μ_{JT} and the speed of sound v using the computed properties and required conversions.

For Problem B, the venting process is modeled as ideal-gas behavior with reversible adiabatic (isentropic) relations. Conservation of energy and entropy relations connect tank conditions to the narrow section at critical conditions, and continuity relates mass flow rate to density, velocity, and area to determine the required diameter.

3 Problem A

To begin this problem we would have to find the Z value at 10°C and 830 bar in the table and in order to find it we will use the CHE calculator. We will first copy the table into excel and then use the function fakima as follows:

- We will find Z at a constant temperature first

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 1, X_{\text{needed}}),$$

where 1 means value/interpolation at, the $X_i : X_f$ are the temperature values we have, $Y_i : Y_f$ are the values of Z at each pressure given at 10°C, and the X_{needed} is the value that we need to interpolate at which in this case is the 10°C

So at 800 atm we get,

$$\text{fakima}(0 : 100, 1.798 : 1.625, 1, 10)$$

and then we will do that for every pressure from 0 atm to 1000 atm and then we will use this table to find at 830 bar = 819.1463 atm which give us Z= 1.785362 which is the Z at constant temperature

Which gives us a $Z = 1.786871767$ We will do the same at constant temperature and get a Z value $Z = 1.792799148$

Now that we have all possible values of Z we can take the average to give us

$$Z = 1.787590612$$

New Table with new values						
	0	10	25	50	75	100
0	1	1	1	1	1	1
1	1	1	1	1	1	1
3	0.999	0.99888	0.999	1	1	1.001
5	0.998	0.9984	0.999	1	1.001	1.001
10	0.996	0.9968	0.998	1	1.001	1.002
30	0.959	0.979082909	0.996	1.001	1.005	1.007
50	0.958	0.977978182	0.996	1.004	1.009	1.014
100	0.984	0.992746182	1.004	1.018	1.027	1.033
200	1.036	1.045146182	1.057	1.072	1.082	1.089
300	1.132	1.138464	1.146	1.154	1.16	1.163
400	1.256	1.255096	1.254	1.253	1.25	1.247
600	1.524	1.51174	1.495	1.471	1.45	1.432
800	1.798	1.760754323	1.723	1.697	1.658	1.625
819.146311	1.823950415	1.787590612	1.746072248	1.718635332	1.677995054	1.643559081
1000	2.067	2.033573913	1.988	1.923	1.867	1.819

Figure 3: Full table with correct Z value at 830 and 10°C

Using our Z values we can make a density table where $\rho = P/RT$ to get the following table of densities in Kmol/m^3 :

Density, Kmol/m^3								
P(bar)	T(K)							
	273.15	283.15	298.15	323.15	348.15	373.15	To calculate density	
	0	0	0	0	0	0	$\rho = P/RTZ$	
1.01325	0.044618	0.043042	0.040876	0.037714	0.035006	0.032661	conversion factors	
3.03975	0.133987	0.12927	0.122752	0.113142	0.105017	0.097884	1 bar*m ³ ---->KJ	
5.06625	0.223535	0.215554	0.204586	0.18857	0.174854	0.163139	R, KJ/Kmol*m ³	
10.1325	0.447967	0.431799	0.409582	0.37714	0.349708	0.325953	$\rho = P*100/RTZ$	
30.3975	1.395751	1.318839	1.231214	1.130289	1.04495	0.973005		
50.6625	2.32868	2.200548	2.052024	1.878187	1.734679	1.610479		
101.325	4.5343	4.335626	4.071347	3.704714	3.408551	3.161716		
202.65	8.61342	8.236505	7.734403	7.036191	6.470576	5.99826		
303.975	11.82443	11.34206	10.70061	9.804329	9.053228	8.424899		
405.3	14.2094	13.71744	13.0387	12.03958	11.20186	10.47651		
607.95	17.56595	17.083	16.40521	15.383	14.48516	13.68458		
810.6	19.85206	19.55606	18.97914	17.77913	16.89062	16.07903		
830	20.03798	19.72348	19.17658	17.97547	17.08878	16.27793		
1013.25	21.58564	21.16557	20.56153	19.61206	18.74977	17.95521		

Figure 4: Density Table

Once we have that we can start calculating the unknown variables.

3.1 A.1 Calculation of Beta, K⁻¹

$$\beta = \frac{-1}{\rho} \times \left(\frac{d\rho}{dT} \right)_P$$

where,

$$\left(\frac{d\rho}{dT} \right)_P$$

is the slope and to find it we will use the fakima again as follows

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 3, X_{\text{needed}}),$$

where 3 is the first derivative, the X values are the temperatures, the Y values are the densities at 830 bar and the X_needed is 10°C. For example

$$= \text{fakima}(273.15 : 373.15, 20.038 : 16.278, 3, 283.15)$$

To give us a slope or $\left(\frac{d\rho}{dT} \right)_P = -0.03296258$.

Now we plug everything in to find Beta,

$$\beta = -1/(19.72348) \times -0.03296258$$

$$\boxed{= 0.00167124\text{K}^{-1}}$$

3.2 A.2 Calculation of Zeta, K⁻¹

$$\zeta = \frac{1}{P} \times \frac{dP}{dT}_V$$

Since we don't have a volume we know that $V = \frac{1}{\rho}$ which means

$$\zeta = \frac{1}{P} \left(\frac{\partial P}{\partial T} \right)_{\rho}$$

To find $\left(\frac{dP}{dT} \right)_{\rho} \rightarrow d\rho = \left(\frac{d\rho}{dT} \right)_P dT + \left(\frac{d\rho}{dP} \right)_T dP$ now solve for

$$\left(\frac{\partial P}{\partial T} \right)_{\rho} = - \frac{\left(\frac{\partial \rho}{\partial T} \right)_P}{\left(\frac{\partial \rho}{\partial P} \right)_T}, \text{ so,}$$

$$\zeta = \frac{1}{P} \left(- \frac{\left(\frac{\partial \rho}{\partial T} \right)_P}{\left(\frac{\partial \rho}{\partial P} \right)_T} \right)$$

We already found $\left(\frac{d\rho}{dT} \right)_P = -0.03296258$ now to find $\left(\frac{d\rho}{dP} \right)_T$ we will still use fakima in the CHE calculator as follows,

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 3, X_{\text{needed}}),$$

where 3 is the first derivative, the X values are the pressures, the Y values are the densities at 10°C and the X_needed is 830 bar. For example

$$= \text{fakima}(0 : 1013.25, 0 : 21.16557, 3, 830)$$

To give us a slope or $\left(\frac{d\rho}{dP} \right)_T = 0.008002965$

Now plug everything into the equation to get Zeta,

$$\zeta = 1/(830) * (-0.008002965 / -0.03296258)$$

$$\boxed{= 0.00492405 \text{K}^{-1}}$$

Finding Beta			
Beta= $-1/p * dp/dT)_P$			
finding $dp/dT)_P$		-0.03296258	
Beta=		0.001671236	
Finding Zeta			
$zeta=1/P * dP/dT)_V$			
which also means			
$Zeta=1/P * dP/dT)_P$			
Fidning $dP/dT)_P$			
$dp =$		$dp/dT)_P dT + dp/dP)_T dP$	
density is constant so $dp=0$			
solve for $dP/dT)_P$			
$dP/dT)_P = (-dp/dT)_P / (dp/dP)_T$			
$zeta=1/P * (-dp/dT)_P / (dp/dP)_T$			
Now find each derivative			
$dp/dT)_P$		-0.03296258	
$dp/dP)_T$		0.008002965	
zeta=		0.004962405	

Figure 5: Beta and Zeta Excel Calculations

3.3 A.3 Calculation of Enthalpy, $\frac{KJ}{Kmol}$

We know that the formula for enthalpy is

$$dH = C_p dT + (1 - \beta T) V dP$$

We also know that T is constant and $V = \frac{1}{\rho}$ so that gives us

$$dH = (1 - \beta) \frac{1}{\rho} dP$$

At 0 bar we know that it is ideal gas so we can use the HIG function in the CHE calculator to find the enthalpy at 0 bar and 10°C to get $-0.43776 \frac{KJ}{Kmol}$

So now we will need to find β at each pressure and in order to do that we will need to find the slope $\left(\frac{d\rho}{dT}\right)_P$ for each pressure at 10°C to get the following table

	T(K)	283.15
Pressure (bar)	Slope or $dp/dT)_P$	
0	0	
1.01325	-0.00015	
3.03975	-0.00046	
5.06625	-0.00077	
10.1325	-0.00156	
30.3975	-0.00676	
50.6625	-0.01137	
101.325	-0.0189	
202.65	-0.03587	
303.975	-0.04582	
405.3	-0.04751	
607.95	-0.04699	
810.6	-0.03387	
830	-0.03296	
1013.25	-0.04126	

Figure 6: $\left(\frac{d\rho}{dT}\right)_P$ Calculation

Now that we have the slope at each pressure we can find β at each pressure and we calculate that just like we calculated the Beta in problem 1 and once we do that we will get

	T(K)	283.15
Pressure (bar)	Beta	
0	0	
1.01325	-0.00015	
3.03975	-0.00046	
5.06625	-0.00077	
10.1325	-0.00156	
30.3975	-0.00676	
50.6625	-0.01137	
101.325	-0.0189	
202.65	-0.03587	
303.975	-0.04582	
405.3	-0.04751	
607.95	-0.04699	
810.6	-0.03387	
830	-0.03296	
1013.25	-0.04126	

Figure 7: Beta Calculation

Now we can do the integrand at each pressure which is $(1 - \beta) * 1/\rho$ at 10°C and we will get the following table

Now we will find the integrand for each pressure at 10°C		
3	Pressure (bar)	Integrand
3		0
0		1.01325 0.00788
1		3.03975 0.01121
2		5.06625 -0.05104
3		10.1325 -0.05182
4		30.3975 -0.34143
5		50.6625 -0.2102
6		101.325 -0.05397
7		202.65 -0.0283
8		303.975 -0.01268
9		405.3 0.00141
0		607.95 0.01295
1		810.6 0.02606
2		830 0.02671
3		1013.25 0.02117

Figure 8: Enthalpy Integrand Calculation

Now that we have the integrand at each pressure we will be using the fakima function in excel to find the enthalpy and we do that as follows

$$= \text{fakima}(X_i : X_f, Y_i : Y_f, 5, \text{lower limit}, \text{higher limit}),$$

where 5 is the integral, the X values are the pressures, the Y values are the integrand at each pressure at 10°C, the lower limit is 0 and the higher limit is 830 bar. For example

$$= \text{fakima}(0 : 1013.25, 0 : 0.026709, 5, 0, 830)$$

And that will give us

$$H = -16.86897077 \frac{\text{bar} \cdot \text{m}^3}{\text{Kmol}}$$

to convert to KJ we will multiply by

$$100 \frac{\text{KJ}}{\text{bar} \cdot \text{m}^3}$$

to get

$$H = -1686.897077 \frac{\text{KJ}}{\text{Kmol}}$$

and then now we can convert to $\frac{\text{KJ}}{\text{mol}}$ to get

$$H = -1.686897077 \frac{\text{KJ}}{\text{mol}}$$

and now to get the total enthalpy we will add the enthalpy at 0 bar and the enthalpy that we integrated to give us:

$$H_{\text{tot}} = -2.124659675 \frac{\text{KJ}}{\text{mol}}$$

5	Now we can find the integral, bar*m^3/kmol	
6		-16.86897077
7	Convert to KJ/Kmol	
8		-1686.897077
9	Convert to KJ/mol	
0		-1.686897077
1		
2	total enthalpy, HIG at 0 bar and the integral, KJ/mol	
3		-2.124659675
4		
5		

Figure 9: Enthalpy Calculation

3.4 A.4 Calculation of Entropy, $\frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$

We know the formula for entropy is

$$ds = C_p \left(\frac{dT}{T} \right) - \beta V dP$$

We know that T is constant and $V = \frac{1}{\rho}$ so the entropy formula becomes

$$ds = -\frac{\beta}{\rho} dP$$

Since we already calculated the Beta at each pressure and at constant temperature already for the previous question we now can do the integrand at each pressure which is just

$$-\frac{\beta}{\rho}$$

and ρ is at constant temperature (10°C), once we do that we get the following table

Pressure (bar)	Integrand
0	0
1.01325	-0.081933936
3.03975	-0.027620856
5.06625	-0.016750947
10.1325	-0.011861931
30.3975	-0.003916546
50.6625	-0.001980484
101.325	-0.001004446
202.65	-0.00049046
303.975	-0.000305358
405.3	-0.000200525
607.95	-0.000101377
810.6	-8.54587E-05
830	-9.88303E-05
1013.25	0

Figure 10: Entropy Integrand Calculation

Now with integrand at each pressure we can take the integral the same way we did for the enthalpy using fakima function in the CHE calculator and once we do that we get

$$S = -0.741618391 \frac{\text{bar} \cdot \text{m}^3}{\text{Kmol} \cdot \text{K}}$$

to get it in $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$ we multiply by

$$100 \frac{\text{KJ}}{\text{bar} \cdot \text{m}^3}$$

to get

$$S = -74.1618391 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

also we know that the entropy at 0 bar should be ideal gas but since the function in the

calculator doesn't provide an entropy at 0 bar we will take it at 1 bar to get

$$SIG = -1.397044779 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

and to get the total entropy we will add the SIG and the S calculated by the integral to get

$$S_{\text{tot}} = -75.55888391 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

Now we can find the integral, bar*m^3/kmol*K	
-0.741618391	
Convert to KJ/Kmol*K	
-74.16183914	
at P=0 bar at 10 C it is ideal gas so SIG, KJ/Kmol K=	
-1.397044779	
Total Entropy	
-75.55888391	

Figure 11: Entropy Calculation

3.5 A.5-6 Calculation of C_p and C_v , $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

We know that

$$C_p - C_v = \left(\frac{T}{\rho^2} \right) \left(\frac{\left(\left(\frac{\partial P}{\partial T} \right)_\rho \right)^2}{\left(\frac{\partial P}{\partial \rho} \right)_T} \right)$$

And

$$\left(\frac{\partial P}{\partial T}\right)_\rho = -\frac{\left(\frac{\partial \rho}{\partial T}\right)_P}{\left(\frac{\partial \rho}{\partial P}\right)_T}$$

We have previously calculated $\left(\frac{\partial \rho}{\partial T}\right)_P = -0.03296258$ and $\left(\frac{\partial \rho}{\partial P}\right)_T = 0.008002965$ so therefore

$$\left(\frac{\partial P}{\partial T}\right)_\rho = -\frac{\left(\frac{\partial \rho}{\partial T}\right)_P}{\left(\frac{\partial \rho}{\partial P}\right)_T} = -\left(\frac{-0.03296258}{0.008002965}\right)$$

to give us

$$\left(\frac{\partial P}{\partial T}\right)_\rho = 4.118795909$$

Then we can calculate $C_p - C_v$ such that

$$C_p - C_v = \left(\frac{283.15}{(19.72347533)^2}\right) \left(\frac{(4.118795909)^2}{0.008002965}\right)$$

to give us

$$C_p - C_v = 0.098819153 \frac{\text{m}^3 \cdot \text{bar}}{\text{Kmol} \cdot \text{K}}$$

and now we convert it to $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$ to give us

$$C_p - C_v = 9.881915289 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

3.5.1 A.5 Calculation of C_v , $\frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

$$dU = C_v dT + (\zeta T - 1) P dV \quad \zeta = \frac{1}{P} \left(\frac{\partial P}{\partial T}\right)_\rho$$

Using the Maxwell equations

$$\left(\frac{\partial C_v}{\partial V}\right)_T = -\frac{\partial}{\partial T} [(\zeta T - 1) P]_V$$

$$\frac{\partial}{\partial T} \left[\left(1 + \frac{\partial P}{\partial T} \right) T \right]_V - \frac{\partial}{\partial T} \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right]_V$$

$$T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho + \left(\frac{\partial P}{\partial T} \right)_\rho - \left(\frac{\partial P}{\partial T} \right)_\rho \Rightarrow \boxed{\left(\frac{\partial C_v}{\partial V} \right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho}$$

$$\int dC_v = \int T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho dV \Rightarrow C_v = C_v^{IG} + \int_\infty^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho dV$$

We will need to evaluate the integral in terms of density such as

$$C_{vIG} + \int_0^\rho \left(T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho \right) \left(\frac{d\rho}{-\rho^2} \right)$$

and we can get the CvIG by using the CHE calculator to get

$$C_{vIG} = 20.87745872$$

and now we will evaluate

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \frac{\rho_{TT}\rho_P^2 - 2\rho_T\rho_{TP}\rho_P + \rho_T^2\rho_{PP}}{\rho_P^3}$$

Definitions:

$$\rho_T = \left(\frac{\partial \rho}{\partial T} \right)_P \quad \rho_P = \left(\frac{\partial \rho}{\partial P} \right)_T$$

$$\rho_{TT} = \left(\frac{\partial^2 \rho}{\partial T^2} \right)_P \quad \rho_{PP} = \left(\frac{\partial^2 \rho}{\partial P^2} \right)_T$$

$$\rho_{TP} = \frac{\partial^2 \rho}{\partial T \partial P} = \frac{\partial}{\partial P} \left(\frac{\partial \rho}{\partial T} \right)$$

To calculate each derivative we will use the function fakima just the same way we have used it to get previous slopes or derivatives all of those calculations were done in excel using the fakima function so I will be putting tables to show the numbers that I have got for each derivative at each point.

dp/dT)_P						
	T(K)					
Pressure (bar)	273.15	283.15	298.15	323.15	348.15	373.15
0	0	0	0	0	0	0
1.01325	-0.000164182	-0.000151959	-0.000136839	-0.000116469	-0.000100258	-8.65541E-05
3.03975	-0.000490192	-0.000455881	-0.000415275	-0.0003512	-0.000301208	-0.000265535
5.06625	-0.000831561	-0.000769645	-0.000693053	-0.000591812	-0.000505822	-0.00042857
10.1325	-0.001684583	-0.00155911	-0.001407104	-0.001186445	-0.001012434	-0.000876678
30.3975	-0.008616006	-0.00675506	-0.00449194	-0.003556301	-0.003125438	-0.002609909
50.6625	-0.01426904	-0.011366479	-0.007820593	-0.005992168	-0.005268401	-0.004581792
101.325	-0.020991864	-0.01889531	-0.01630805	-0.012975499	-0.010685854	-0.00888683
202.65	-0.03980046	-0.035869158	-0.031017156	-0.024758261	-0.020434029	-0.017026655
303.975	-0.050973553	-0.045818237	-0.039409626	-0.032456032	-0.027383289	-0.022677739
405.3	-0.051169254	-0.047508779	-0.043244508	-0.03647601	-0.030858905	-0.026766579
607.95	-0.049849501	-0.046989922	-0.043533249	-0.038277089	-0.033730332	-0.030078584
810.6	-0.025170929	-0.03386758	-0.042425374	-0.038579007	-0.033073059	-0.030925507
830	-0.028945012	-0.032962578	-0.03976007	-0.038077752	-0.033023326	-0.030916777
1013.25	-0.042874793	-0.041257568	-0.039508136	-0.036381129	-0.032966894	-0.030427721

Figure 12: $\left(\frac{\partial \rho}{\partial T} \right)_P$

(d^2p/dT^2)_P						
	T(K)					
Pressure (bar)	273.15	283.15	298.15	323.15	348.15	373.15
0	0	0	0	0	0	0
1.01325	1.51923E-06	1.02241E-06	8.5331E-07	6.57316E-07	4.50872E-07	6.45475E-07
3.03975	4.26486E-06	3.11556E-06	2.28639E-06	2.29328E-06	9.52041E-07	1.90179E-06
5.06625	7.6956E-06	5.17897E-06	4.47752E-06	3.48494E-06	2.75561E-06	3.42452E-06
10.1325	1.55953E-05	1.0923E-05	8.60431E-06	7.48434E-06	4.07458E-06	6.78584E-06
30.3975	0.000182683	6.36168E-05	3.43332E-05	-2.16796E-07	1.81907E-05	2.30516E-05
50.6625	0.000292975	0.000113165	6.18283E-05	2.5431E-06	1.71756E-05	3.77532E-05
101.325	0.000255337	0.000165706	0.000127653	8.77824E-05	5.10684E-05	9.28536E-05
202.65	0.000479131	0.000311329	0.000240576	0.000166134	9.73424E-05	0.000175248
303.975	0.000610928	0.000367374	0.000297754	0.000173058	0.000163584	0.00021286
405.3	0.000451864	0.000335159	0.000245671	0.000262765	0.000115396	0.00021199
607.95	0.000360789	0.000260689	0.000214245	0.000203553	0.000117497	0.000174643
810.6	-0.000918708	-0.000696377	-0.00164567	0.000288785	-2.55847E-05	0.000197389
830	-0.000699485	-0.000492639	-0.00212281	0.00022209	-2.70155E-05	0.000195539
1013.25	0.000197566	0.000161875	0.000116865	0.000180348	8.11572E-05	0.000121977

Figure 13: $\left(\frac{\partial^2 \rho}{\partial T^2} \right)_P$

dp/dP)_T							
P(bar)	T(K)						
		273.15	283.15	298.15	323.15	348.15	373.15
0	0.044001006	0.042443193	0.040311504	0.03722081	0.034548053	0.032257582	
1.01325	0.04406236	0.042530608	0.04038722	0.03722081	0.034548053	0.032197205	
3.03975	0.044133182	0.042561356	0.04039372	0.03722081	0.034548053	0.032192037	
5.06625	0.044192392	0.04258025	0.040398273	0.03722081	0.03447705	0.032196601	
10.1325	0.044624206	0.04299603	0.040517239	0.03722081	0.034480594	0.032112118	
30.3975	0.046405273	0.043655346	0.040539851	0.037149128	0.034260854	0.03183572	
50.6625	0.045579386	0.043417012	0.040496815	0.03684146	0.033947868	0.031329083	
101.325	0.042794944	0.041602997	0.039540277	0.035630679	0.03254934	0.030168696	
202.65	0.037800896	0.035862343	0.03357801	0.030786725	0.028341728	0.026316079	
303.975	0.02719621	0.026797552	0.026078314	0.024690277	0.023407023	0.022179763	
405.3	0.019305533	0.019200622	0.019112564	0.019115671	0.018717655	0.018128988	
607.95	0.012315338	0.013715621	0.013799865	0.012920142	0.012947111	0.01286488	
810.6	0.009883869	0.009156041	0.011189627	0.010465867	0.010561576	0.010589064	
830	0.008901871	0.008002965	0.008891705	0.009419887	0.009535481	0.009607381	
1013.25	0.007876953	0.007489403	0.006248006	0.008336223	0.008489003	0.008602899	

Figure 14: $\left(\frac{\partial \rho}{\partial P}\right)_T$

d^2p/dP^2)_T						
Pressure (bar)	T(K)					
	273.15	283.15	298.15	323.15	348.15	373.15
0	7.46544E-05	3.89868E-05	2.98906E-05	2.28718E-18	-1.55689E-17	-2.38351E-05
1.01325	4.20854E-05	2.81343E-05	3.8421E-05	3.07966E-18	-3.15657E-18	-3.06529E-05
3.03975	0.000105522	2.96209E-05	-3.86839E-05	-2.73859E-18	-0.000185391	2.27131E-05
5.06625	-4.3609E-05	-4.17655E-05	2.9809E-05	1.44453E-19	4.18164E-05	-3.72867E-05
10.1325	0.0004594	0.00016468	5.80734E-06	-9.43938E-06	-2.95658E-05	-2.67902E-05
30.3975	-2.7685E-05	-1.98209E-05	-6.42488E-06	-4.16615E-05	-3.58397E-05	-6.21483E-05
50.6625	-0.000132131	-7.92517E-05	-3.78555E-05	-4.55985E-05	-5.23497E-05	-3.82844E-05
101.325	-5.16643E-05	-7.05113E-05	-8.29785E-05	-6.7322E-05	-5.48905E-05	-5.27004E-05
202.65	-0.000152527	-0.000129758	-0.000106822	-8.49848E-05	-7.1534E-05	-5.85191E-05
303.975	-6.0883E-05	-4.86794E-05	-4.03394E-05	-4.57059E-05	-3.78116E-05	-3.44434E-05
405.3	-1.22036E-05	-2.26361E-05	-2.15903E-05	-1.63437E-05	-1.75365E-05	-1.60981E-05
607.95	-6.62483E-06	2.31246E-07	-6.76375E-06	-8.23237E-06	-8.34707E-06	-8.60346E-06
810.6	8.18933E-06	-4.38459E-05	-7.62327E-05	9.56776E-07	-1.65851E-06	-2.7404E-06
830	-3.75197E-06	1.23752E-06	-1.48239E-05	-4.18282E-06	-4.01393E-06	-3.91692E-06
1013.25	-7.43404E-06	-6.84255E-06	-1.40296E-05	-7.64435E-06	-7.40738E-06	-7.04605E-06

Figure 15: $\left(\frac{\partial^2 \rho}{\partial P^2}\right)_T$

d/dP(dp/dT)						
P(bar)	T(K)					
	273.15	283.15	298.15	323.15	348.15	373.15
0	-0.000162616	-0.000149971	-0.000133875	-0.000114504	-9.88405E-05	-8.39732E-05
1.01325	-0.000161881	-0.000149972	-0.000133875	-0.000114504	-9.88405E-05	-8.39732E-05
3.03975	-0.000167971	-0.000149985	-0.000133875	-0.000114504	-9.88405E-05	-8.39724E-05
5.06625	-0.000168449	-0.000154876	-0.000133875	-0.000114504	-9.88405E-05	-8.39733E-05
10.1325	-0.00016859	-0.000159191	-0.000133875	-0.000114504	-9.88405E-05	-8.3973E-05
30.3975	-0.000307795	-0.000240241	-0.000133875	-0.000114504	-9.88405E-05	-8.3973E-05
50.6625	-0.000199425	-0.000179871	-0.000133875	-0.000114504	-9.88405E-05	-8.39729E-05
101.325	-0.000167629	-0.000158677	-0.000133875	-0.000114504	-9.88405E-05	-8.39729E-05
202.65	-0.000160894	-0.000154462	-0.000133875	-0.000114504	-9.88405E-05	-8.39729E-05
303.975	-1.28478E-05	-3.43933E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.3973E-05
405.3	2.15971E-06	-5.76883E-06	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
607.95	9.50943E-06	3.46067E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
810.6	-3.41763E-05	5.74475E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
830	-0.000108321	3.15245E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05
1013.25	-1.67526E-05	-9.12237E-05	-0.000133875	-0.000114504	-9.88405E-05	-8.39731E-05

Figure 16: $\frac{\partial^2 \rho}{\partial T \partial P}$

Now that we have all the numbers needed to solve for the integral of Cv we will now do the integrand which is

$$= \left(\frac{\partial^2 P}{\partial T^2} \right)_{\rho} \left(\frac{-1}{\rho^2} \right) d\rho$$

Once we plug everything in at 10°C and the densities at 10°C we get the following integrand table

Since we are using the density as the integral limits we will need to use the densities at 10 C because it is Cv(T,p)			
Densities, Km ³ /m ³	Integrand= T*(d ² P/dT ²) _p *(-1/p ²))dp		
0	0		
0.04304176	0.00062077		
0.129270064	0.00013226		
0.215553688	0.000219119		
0.431799363	5.05849E-05		
1.318839091	0.000147575		
2.200548093	-6.44672E-05		
4.335625889	-8.98241E-06		
8.236505327	5.23907E-05		
11.34206101	-3.11707E-05		
13.71744009	-4.65063E-05		
17.08300123	-0.000125051		
19.556055	0.000248825		
19.72347533	6.80822E-05		
21.16557465	0.000313205		

Figure 17: Cv Integrand Calculation

Once we have that we can now integrate using fakima function where the densities from 0 bar to 830 bar at 10°C are the limits or the X values in fakima and the Y values are the integrand that we just calculated and multiply it by the temperature in Kelvin= 283.15 K and multiply by 100 KJ to get rid of the bar · m³ Once we do that we get

$$C_v = 2.44761456 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

and to get the total C_v we will add the C_{vIG} to the integral that we evaluated to get

$$C_v = 20.87745872 + 2.44761456$$

to give us

$$C_v = 23.32511018 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

3.5.2 A.6 Calculation of $C_p, \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

To find C_p we will use

$$C_p - C_v = 9.881915289 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

and we just calculated

$$C_v = 23.32511018 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

so now we just solve for

$$C_p = 9.881915289 + 23.32511018$$

to give us

$$C_p = 33.20702547 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

Integral*100 to convert to KJ/Kmol*K	
2.447651456	
Cv=	23.32511018
Cp=Cv+BetaTZetaP/p	33.20702547

Figure 18: C_p and C_v calculation

3.6 A.7 Calculation of the Joule-Thomson Coefficient in $\frac{\text{K}}{\text{bar}}$

We know that the formula for the Joule Thomson Coefficient is

$$\mu_{JT} = \frac{1}{C_p} (\beta T - 1) \frac{1}{\rho}$$

and multiply by $100 \frac{\text{KJ}}{\text{bar} \cdot \text{m}^3}$ to get it in

$$\frac{\text{K}}{\text{bar}}$$

$$\mu_{JT} = \frac{1}{33.207} \left(\frac{(0.00161)(283.15) - 1}{19.723} \right) \times 100$$

To give

$$\mu_{JT} = -0.080431055 \frac{\text{K}}{\text{bar}}$$

3.7 A.8 Calculation of Speed of Sound, $\frac{\text{m}}{\text{s}}$

we know that the formula for the speed of sound is

$$v^2 = \frac{1}{M_w} \gamma \left(\frac{\partial P}{\partial \rho} \right)_T$$

we know that:

$$M_w = 28.013 \frac{\text{kg}}{\text{Kmol}}$$

$$\left(\frac{\partial P}{\partial \rho} \right)_T \text{ was previously calculated which is } 124.9536948 \frac{\text{bar} \cdot \text{m}^3}{\text{Kmol}}$$

$$\gamma = \frac{C_p}{C_v} = \frac{33.207}{23.325}, \gamma = 1.423659962$$

we will need to Convert to get units of $\frac{\text{m}}{\text{s}}$ so we will use

$$\frac{100 \text{ KJ}}{1 \text{ bar} \cdot \text{m}^3} \quad \text{and} \quad \frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}}$$

This will give us

$$v = \sqrt{\left(\frac{1}{28.013} \right) (1.42)(124.951) \left(\frac{100 \text{ KJ}}{1 \text{ bar} \cdot \text{m}^3} \right) \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}} \right)}$$

Which will give us,

$$v = 796.8890813 \frac{\text{m}}{\text{s}}$$

4 Problem B

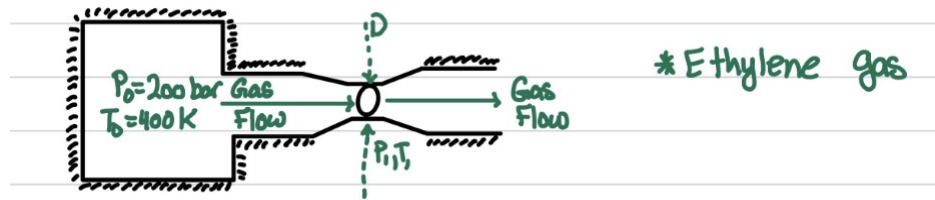


Figure 19: Hand Drawn Diagram

- Pressure drops, velocity increase until reaches speed of sound
- Volume = 20 m³
- Critical Condition
- Reversible and adiabatic
- Behaves as ideal gas
- KE $\neq$ 0

4.1 B.1. Initial mass of ethylene in the tank, kg:

MW : 28.0540 kg/kmol

$$\rho = \frac{P}{RT}$$

$$m = \rho V$$

$$R = 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Conversion factor: 100 kJ = 1 bar · m³

$$\rho = \frac{(200 \text{ bar})}{\left(8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}\right) (400 \text{ K})} = 0.0601395 \frac{\text{bar} \cdot \text{kmol}}{\text{kJ}} \times 100 \frac{\text{kJ}}{1 \text{ bar} \cdot \text{m}^3} = 6.013952 \frac{\text{kmol}}{\text{m}^3}$$

$$m = 6.013952 \frac{\text{kmol}}{\text{m}^3} \cdot 20 \text{ m}^3 \cdot 28.0540 \frac{\text{kg}}{\text{kmol}}$$

$$\boxed{m = 3,374.31 \text{ kg}}$$

4.2 B.2 The Temperature and Pressure at the narrow pipe section:

- $\dot{m} = 5000 \text{ kg/hr} \rightarrow \text{Critical Condition}$
- corresponds to speed of sound
- Reversible and Adiabatic is an isentropic Process

$$-\Delta(mS) + \sum_i \frac{\dot{Q}_i}{T_i} + \dot{S}_{gen} = 0$$

$\dot{Q} = 0$, adiabatic and $\dot{S}_{gen} = 0$, reversible so $\Delta S = 0$, isentropic

we have :
$$\int_{T_1}^{T_2} C_p^{IG} \frac{dT}{T} = R \ln\left(\frac{P_2}{P_1}\right)$$

and we know $\gamma = \frac{C_p}{C_v}$ and $\frac{R}{C_p} = \frac{C_p - C_v}{C_p} = 1 - \frac{1}{\gamma}$

and $C_p^{IG} = \text{Constant} = C_p$ so,

$$\begin{aligned}
C_p \ln\left(\frac{T_2}{T_1}\right) &= R \ln\left(\frac{P_2}{P_1}\right) \\
\ln\left(\frac{T_2}{T_1}\right) &= \frac{R}{C_p} \ln\left(\frac{P_2}{P_1}\right) \Rightarrow \ln\left(\frac{T_2}{T_1}\right) = \left[1 - \frac{1}{\gamma}\right] \ln\left(\frac{P_2}{P_1}\right) \\
\Rightarrow \ln\left(\frac{T_2}{T_1}\right) &= \ln\left[\left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}}\right]
\end{aligned}$$

$$\frac{T_2}{T_1} = \left[\frac{P_2}{P_1}\right]^{1-\frac{1}{\gamma}} \quad (1)$$

Now,

$$\frac{dE_{CV}}{dt} = \dot{Q} - \dot{W} + \dot{m}_{in} \left[h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right] - \dot{m}_{out} \left[h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right]$$

It's a steady flow , adiabatic, no work, and PE=0 , so,

$$0 = 0 - 0 + \dot{m} \left[(h_{in} - h_{out}) + \left(\frac{V_{in}^2}{2} - \frac{V_{out}^2}{2} \right) \right]$$

and since there's no velocity at the beginning , $V_1 = 0$ m/s

$$0 = \dot{m} \left[(h_{in} - h_{out}) - \frac{V_{out}^2}{2} \right]$$

$$h_{in} = h_{out} + \frac{V_{out}^2}{2} \quad (2)$$

we know that $C_p = \left(\frac{\partial h}{\partial T} \right)_p \Rightarrow dh = C_p dT$

$$C_p T_{in} = C_p T_{out} + \frac{V_{out}^2}{2} \quad (3)$$

In the book we know the equation for the speed of sound is

$$v^2 = \frac{1}{M_w} \left(\gamma \frac{\partial P}{\partial \rho} \right)_T$$

and after derivations we get

$$v = \sqrt{\frac{\gamma R T}{M_w}}, \text{ where } \left(\frac{\partial P}{\partial \rho} \right)_T = R T, \text{ (pg.114)}$$

So since the velocity is equal to the speed of sound

$$\text{So } \Rightarrow C_p T_{in} = C_p T_{out} + \frac{\left(\sqrt{\frac{\gamma R T_{out}}{M_w}} \right)^2}{2}$$

$$\Rightarrow C_p T_{in} = C_p T_{out} + \frac{\gamma R T_{out}}{2 M_w}$$

$$\Rightarrow T_{in} = T_{out} + \frac{R}{C_p} \frac{\gamma T_{out}}{2 M_w}$$

$$T_{in} = T_{out} + \left(1 - \frac{1}{\gamma} \right) \frac{\gamma T_{out}}{2 M_w} \quad (4)$$

To find γ we will use the CHE Calculator to find C_p so :

$$\Rightarrow \text{CPIG(CompID , T)}$$

=> CPIG((Comp),(400-273.15)) , where Comp is ethylene and the T is temperature in °C

This gives $C_p = 54.2060 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$

We Know $C_p - C_v = R$

$$C_v = C_p - R \Rightarrow C_v = 54.2060 - 8.314 = 45.8920 \frac{\text{KJ}}{\text{Kmol} \cdot \text{K}}$$

$$\gamma = \frac{C_p}{C_v} = \frac{54.2060}{45.8920} = 1.1812$$

Finding T_{out} first by Plugging everything in (4):

$$400K = T_{out} + \left(1 - \frac{1}{1.1812}\right) \frac{1.1812 T_{out}}{2 \left(28.0540 \frac{\text{kg}}{\text{kmol}}\right)} \cdot 28.0540 \frac{\text{kg}}{\text{kmol}}$$

$$400k = T_{out} + 0.0906T_{out}$$

$$400k = 1.0906T_{out}$$

$$\boxed{T_{out} = 366.771 \text{ K}}$$

Now finding P_{out} by Plugging into (1):

$$\frac{T_{out}}{T_{in}} = \left(\frac{P_{out}}{P_{in}} \right)^{1-\frac{1}{\gamma}}$$

$$\frac{366.771}{400} = \left(\frac{P_{out}}{200} \right)^{1-\frac{1}{1.1812}}$$

$$0.917 = \left(\frac{P_{out}}{200} \right)^{0.153}$$

$$0.568 = \frac{P_{out}}{200} \Rightarrow \boxed{P_{out} = 113.63 \text{ bar}}$$

4.3 B.3 The diameter of the narrow section , mm:

We know that :

$$\dot{m} = \rho \cdot v \cdot A \quad \text{where} \quad v = \text{Velocity} , \quad N_{out} = \text{Speed of sound} , \quad N_{out} = \sqrt{\frac{\gamma RT}{M_w}}$$

$$\rho = \text{density} \quad A = \text{Area} \quad \dot{m} = \text{mass flow rate}$$

$$\rho \text{ for ideal gas : } \rho = \frac{PM_w}{RT}$$

$$\rho = \frac{(113.63 \text{ bar})(28.0540 \text{ kg/kmol})}{(8.314 \text{ KJ/kmol.K})(366.771 \text{ K})} \cdot \left(\frac{100 \text{ KJ}}{1 \text{ bar} \cdot \text{m}^3} \right) = 104.54 \text{ kg/m}^3$$

$$N_{out} = \sqrt{\frac{(1.1812)(8.314 \text{ KJ/kmol.k})(366.771 \text{ K})}{28.0540 \text{ kg/kmol}}} \cdot \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ KJ/Kg}} \right) = 358.316 \text{ m/s}$$

$$A = \frac{\dot{m}}{\rho \cdot N_{out}} = \frac{5000 \text{ kg/hr}}{(104.54 \text{ kg/m}^3)(358.316 \text{ m/s})} \cdot \frac{1 \text{ hr}}{3600 \text{ s}} = 3.71 \times 10^{-5} \text{ m}^2$$

Now find the diameter :

$$d = 2\sqrt{\frac{A}{\pi}} = 2\sqrt{\frac{3.71 \times 10^{-5} \text{ m}^2}{\pi}} = 0.00687 \text{ m} \times \frac{10^3 \text{ mm}}{1 \text{ m}}$$

$$\boxed{6.87 \text{ mm}}$$

4.4 B.4 Mass of ethylene in tank when Critical Conditions are reached, kg:

This is Still ideal gas so,

$$P_1 V = m_1 R T_1 \quad \text{and} \quad P_2 V = m_2 R T_2$$

we also know:

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{1-\frac{1}{\gamma}}$$

$$\frac{\frac{P_2 V}{m_2 R}}{\frac{P_1 V}{m_1 R}} = \left(\frac{P_2}{P_1} \right)^{1-\frac{1}{\gamma}}$$

$$\frac{P_2}{m_2} \cdot \frac{m_1}{P_1} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}} \quad \text{we need } \frac{m_2}{m_1} \text{ so take inverse :}$$

$$\frac{m_2}{P_2} \cdot \frac{P_1}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)}$$

$$\Rightarrow \frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)} \cdot \frac{P_2}{P_1} \quad * \quad -1 + \frac{1}{\gamma} + 1 = \frac{1}{\gamma}$$

$$\frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} \Rightarrow m_2 = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} m_1$$

$$\Rightarrow m_2 = \left(\frac{113.63}{200}\right)^{\frac{1}{1.1812}} (3,374.308) = \boxed{2,090.8069 \text{ kg}}$$

4.5 B.5 Mass of ethylene during the venting Process if $P_{gauge}=P_1=2300$

Psig , kg:

$$P_{sys} = P_{gauge} + P_{atm} \Rightarrow 2300 \text{ Psig} + 14.7 \text{ Psi} \Rightarrow 2314.7 \text{ Psia}$$

$$2314.7 \text{ Psia} \times \left(\frac{0.0689 \text{ bar}}{1 \text{ Psia}}\right) = 159.593 \text{ bar}$$

This is Still ideal gas so,

$$P_1 V = m_1 R T_1 \quad \text{and} \quad P_2 V = m_2 R T_2$$

we also know :

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}}$$

$$\frac{\frac{P_2 V}{m_2 R}}{\frac{P_1 V}{m_1 R}} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}}$$

$$\frac{P_2}{m_2} \cdot \frac{m_1}{P_1} = \left(\frac{P_2}{P_1}\right)^{1-\frac{1}{\gamma}} \quad \text{we need } \frac{m_2}{m_1} \text{ so take inverse :}$$

$$\frac{m_2}{P_2} \cdot \frac{P_1}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)}$$

$$\Rightarrow \frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{-\left(1-\frac{1}{\gamma}\right)} \cdot \frac{P_2}{P_1} \quad * \quad -1 + \frac{1}{\gamma} + 1 = \frac{1}{\gamma}$$

$$\frac{m_2}{m_1} = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} \Rightarrow m_2 = \left(\frac{P_2}{P_1}\right)^{\frac{1}{\gamma}} m_1$$

Plug everything in:

$$m_2 = \left(\frac{159.593 \text{ bar}}{200 \text{ bar}}\right)^{\frac{1}{1.1812}} \cdot (3374.31 \text{ kg})$$

$$\boxed{m = 2,787.43 \text{ kg}}$$

5 Results Summary

Problem	Quantity	Reported Answer
A.1	β	0.00167124
A.2	ζ	0.00492405
A.3	H_{tot}	$-2.124659675 \frac{\text{KJ}}{\text{mol}}$
A.4	S_{tot}	$-75.55888391 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.5–6	$C_p - C_v$	$9.881915289 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.5	C_v	$23.32511018 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.6	C_p	$33.20702547 \frac{\text{KJ}}{\text{Kmol}\cdot\text{K}}$
A.7	μ_{JT}	$-0.080431055 \frac{\text{K}}{\text{bar}}$
A.8	v	$796.8890813 \frac{\text{m}}{\text{s}}$
B.1	m (initial tank mass)	3,374.31 kg
B.2	T_{out}	366.771 K
B.2	P_{out}	113.63 bar
B.3	d	6.87 mm
B.4	m_2 (venting mass at $P_2 = 113.63$ bar)	2,090.8069 kg
B.5	m_2 (venting mass at $P_{\text{gauge}} = 2300$ Psig)	2,787.43 kg

6 Conclusion

In Problem A, the target state at 10°C and 830 bar was evaluated using interpolation and density-based thermodynamic identities. Akima interpolation (FAKIMA) was used to compute needed slopes/derivatives, construct integrands, and evaluate integrals for enthalpy, entropy, and C_v . From these results, additional properties were determined including C_p , the Joule–Thomson coefficient, and the speed of sound.

In Problem B, ethylene venting was analyzed under ideal-gas and isentropic assumptions to determine the initial tank mass, narrow-section temperature and pressure at the critical condition, the narrow-section diameter, and the mass remaining during the specified venting conditions. Overall, the assignment demonstrates how interpolation, derivative relations, and conservation laws are combined to compute thermodynamic and flow properties for engineering analysis.

7 References

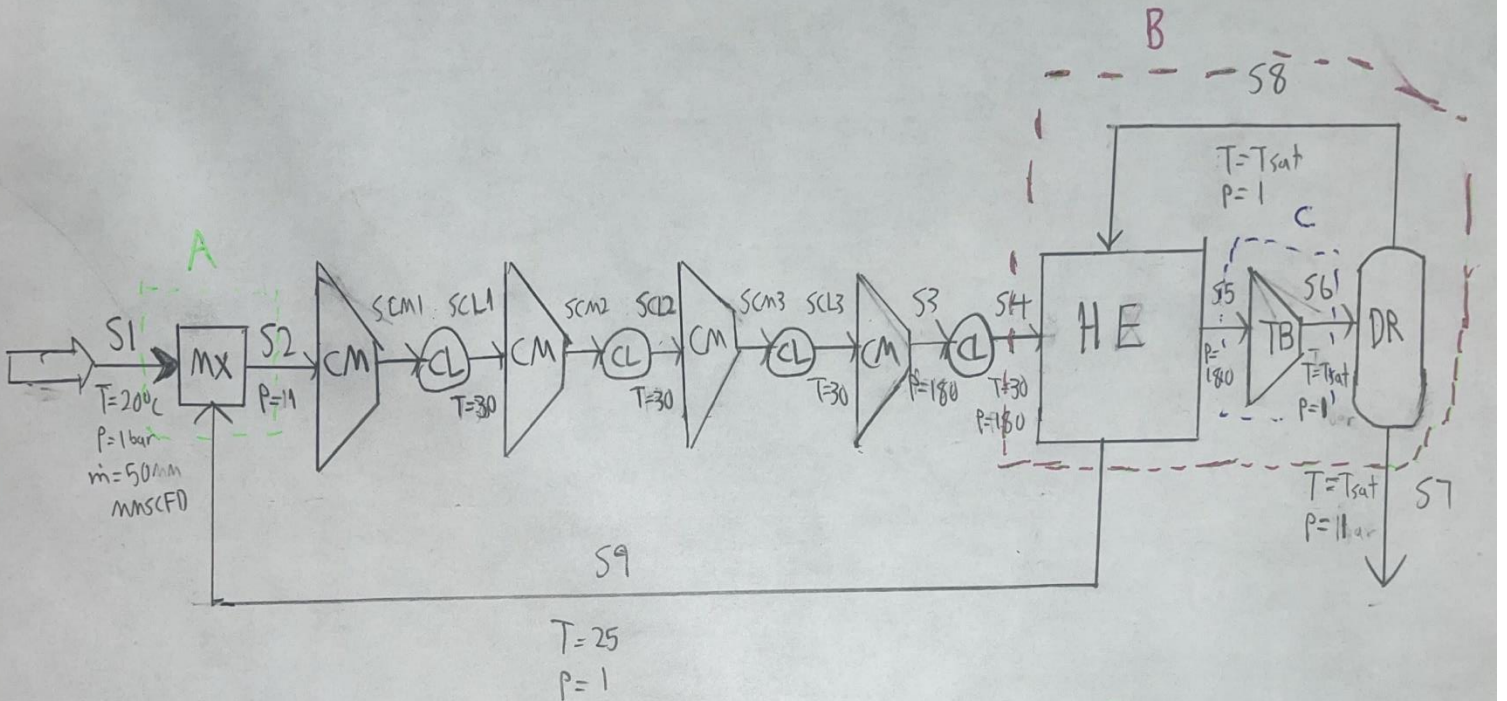
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Diagram of the updated system; includes the four-stage compressor with intercoolers and the turbine. This is known as the Claude Process, not the Linde Process, since it employs a turbine.

CHE 301
Qussay Suleiman

Liquefaction of Nitrogen via the Claude Process



MX = mixer

DR = drum

CM = Compressor

CL = cooler

HE = heat exchanger

TB = Turbine (expander)

System A is around the mixer

System B is around the HE, TB, and DR

System C is around the Turbine

Introduction

This Claude process aims to convert nitrogen vapor into liquid nitrogen while minimizing power consumption. This method is essential in industrial applications where liquid nitrogen is required for cooling and various cryogenic processes. The turbine generates power for the system to offset some of the power consumed by the compressors, and the four-stage compressor reduces energy usage by incrementing pressure changes. While not explicitly stated, we will assume that the process is adiabatic and that the pressure ratio between compressors is equal. Most calculations will be made using the CHE Calculator provided by Dr. Michael Caracotsios. The exact numerical answers can be found on the final page, while only the methods will be explained throughout the report.

Formulas and Balances

Although most of the thermal properties will be solved using the CHE Calculator functions, some balances and formulas will be referenced throughout the report. They will be adjusted depending on the conditions of each system, and are as follows:

- Mass Balance:

Since there is no accumulation anywhere in this process, the mass flow rate of the nitrogen fed into a component and exit in that component will be equal,

$$m_{in} = m_{out}$$

- Energy Balance:

Since the entire system is assumed to be adiabatic, the energy balance is,

$$-\Delta mH + Q + W_s = 0 \rightarrow \Delta mH = W_s$$

The work term is zero for all systems excluding around the compressors and the turbine, and ideal work is used for the ideal turbine stream.

- Entropy Balance:

Again, since the entire system is adiabatic, the entropy balance simplifies as follows,

$$-\Delta mS + \frac{Q}{T_0} + S_{gen} = 0 \rightarrow \Delta mS = S_{gen} \text{ or } S_{in} = S_{out} \text{ for the ideal turbine stream}$$

- Lost Work Balance:

Again, the entire system is adiabatic, so the Lost Work Balance simplifies as follows,

$$LW = -\Delta m(H - T_0S) + Q \left(1 - \frac{T_0}{T}\right) + W_s \rightarrow LW = -\Delta m(H - T_0S) + W_s$$

Like the energy balance, the work term is zero for all systems excluding those around the compressors and the turbine.

- Goal:

The goal is the change in Availability (B) flow for the overall system, $B = H - T_0S$ and $T_0 = 300$ K. It can be written and specified as follows for our process,

$$Goal = -\Delta mB = m_1B_1 - m_7B_7$$

- Thermodynamic Efficiency:

$$\eta = \frac{(Goal) - (LW)}{(Goal)} \text{ when Goal} > 0$$

$$\eta = \frac{+Goal}{Goal - (LW)} \text{ when Goal} < 0$$

Breaking down the process

The process is broken down into fifteen streams. All streams are as shown in the provided images, excluding most of the compressor and cooler streams which are shown in my updated diagram. The acronym “CM” refers to the nitrogen vapor exiting a compressor, and “CL” refers to the nitrogen vapor exiting a cooler, except for the last compressor and cooler. The hand-drawn diagram should be referenced when searching for given data or names for the streams or systems. The individual processes can be categorized in two ways to help with writing balances:

1. Processes with one inlet and one outlet:
This refers to systems where there is only one stream entering and exiting the equipment. All the compressors, coolers, and the turbine are examples of this type of process.
2. Processes with multiple inlets and/or outlets:
This refers to systems where there is more than one input, more than one output, or both. The system around the mixer, heat exchanger, and drum are examples of this type of process.

Identify the T, P, and flow rate of each stream, and guess unknowns

The temperature, pressure, and flow rate of some of the streams are given in the problem. However, those do not include the four-stage compressor with intercoolers, so an updated table that includes vapor fraction will be written. Values in red are initial guesses. The pressure out of each compressor, although not explicitly given, change by a ratio of $\sqrt[4]{180}$ because the pressure out of the final compressor is 180 bar and there are four compressors. Since there is no pressure drop in the coolers, the pressure out of the turbines will equal the pressure out of each subsequent cooler:

Stream No →	Temperature, °C	Pressure, bar	Flow Rate, MMSCFD	Vapor Fraction
1	20	1	50	1
2	23	1	?	1
CM1	?	3.663	?	1
CL1	30	3.663	?	1
CM2	?	13.416	?	1
CL2	30	13.416	?	1
CM3	?	49.142	?	1
CL3	30	49.142	?	1
3	?	180	?	1
4	30	180	?	1
5	-100	180	?	1
6	T ^{sat}	1	?	?
7	T ^{sat}	1	?	0
8	T ^{sat}	1	?	1
9	25	1	3	1

The methods to find the missing data in each column are as follows:

- Molar flow rate:

Although most of the flow rates are unknown, they are easiest to calculate. The flow rate of stream 2, since it's the output of the mixer, will equal the sum of both inputs to the mixer. This can be written as: $m_2 = m_1 + m_9$

The flow rates of every output stream of a single output and input system will equal the flow rate of the input. The categorization for each system can be found in the "Breaking down the process" section. Despite possessing two inputs and two outputs, the same applies to the streams going in and out of the heat exchanger, since the streams do not mix. To be specific, the flow rate of stream 4 will equal that of stream 5, and the flow rate of stream 8 will equal that of stream 9.

The next system, the drum, will have a mass balance similar to the mixer, except it has two outputs rather than two inputs. It can be written as: $m_6 = m_7 + m_8$

To find the flow rate of stream 7, an overall mass balance must be made. Since there is only a singular input and output for the overall system, $m_1 = m_7$

A flow rate for one of the recycle streams (8 or 9), must be guessed to find all other flow rates. For consistency with our in-class problem, I will guess the flow rate,

$m_9 = 3 \frac{\text{kmol}}{\text{s}}$. I chose this flow rate because it is significantly higher than m_1 , which is reasonable because most of the nitrogen does not get converted into a liquid state and is instead recycled. After this guess, all the molar flow rates are known. We convert m_1 from MMSCFD into kmol/s using the CHE calculator function "MMSCFDtoKMOLSEC()", which we do because it simplifies calculations.

- Temperature:

Although there are fewer unknown temperatures compared to unknown flow rates, they are trickier to find. The temperature of the streams out of the compressors can be found using the CHE calculator function "GasCompTemp()" since efficiency is given and everything else is known. The temperature of stream 2 and stream 5 need to be guessed. I guess $T_2 = 23^\circ\text{C}$ since it must be between the temperature of stream 1 and stream 9. I guessed $T_5 = -100^\circ\text{C}$ since it must be between the temperature of stream 3 and stream 9. Although it's not shown on the table, I find the ideal temperature of stream 6 using the CHE calculator function "GasTurbTemp()".

- Vapor Fraction:

All streams are vapor, excluding stream 7 and stream 6. While these are not provided, they are known by the nature of this process. The drum separates stream 6 into separate vapor and liquid streams, where stream 7 is liquid and stream 8 is vapor. Stream 6 is a mixture between vapor and liquid, and can be found using, $vf_6 = \frac{m_8}{m_6}$, which was given in the lecture.

Calculate the Thermal Properties of each stream

We use the Peng-Robinson equation of state. The compressibility factor for both the liquid and vapor states of all streams can be found using the “ZComp()” function of the CHE calculator. With Z found, we can use the “Enthalpy()” and “Entropy()” functions of the CHE calculator to find the enthalpy and entropy for the liquid and vapor states of all streams, and then use the lever rule to find the actual enthalpy and entropy. They can be written as follows:

$$H = X * H^V + (1 - X) * H^L \quad \text{and} \quad S = X * S^V + (1 - X) * S^L$$

However, the lever rule is not necessary for most streams, since only stream 6 is a mixture. The Zcomp for the liquid state is only needed in stream 6 and stream 7 since the other streams are purely vapor. I then multiply the enthalpy and entropy of each stream, which equals their enthalpy flow and entropy flow. The ideal turbine process, however, is written differently. Enthalpy is solved the same way as the actual process. For entropy, I simply set $S_{6(\text{ideal})} = S_5$ directly by setting the Excel cell for $S_{6(\text{ideal})}$ equal to the Excel cell for S_5 . I then use that entropy to solve for the ideal quality of the turbine, by rearranging the lever rule as follows: $X = \frac{S - S^L}{S^V - S^L}$

Identify and solve the equations to solve the unknowns

I made three equations to solve for the three unknowns. The unknowns are shown in the table. The equations are energy balances around systems A, B, and C. They are as follows:

$$m_1 H_1 + m_9 H_9 - m_2 H_2 = 0 \quad (\text{System A})$$

$$m_4 H_4 - m_7 H_7 - m_9 H_9 = 0 \quad (\text{System B})$$

$$m_5 H_5 - m_6 H_6 - W_{s(\text{ideal})} = 0 \quad (\text{System C})$$

I use the Newton Method to set all three equations to zero while changing T_2 , T_5 , and m_9 .

I make a guess for the ideal pressure out of the turbine as well, but that is only used to find the ideal work of the turbine, so I will discuss it in the next section. The final answers for unknowns can be seen on the Excel screenshot.

Calculate Power

Finding power in this Claude Process is significantly harder compared to the Linde Process example we completed in the lecture. We have four compressors and a turbine this time, as opposed to a single compressor. The power for the compressors can be found by using the energy balance found in the second section. We subtract the enthalpy flow of the output stream by the enthalpy flow of the input stream, which can be written as follows:

$$W_{s(\text{comp})} = m \Delta H, \text{ where } W_{s(\text{comp})} > 0 \text{ because compressors consume power}$$

Finding the power of the turbine is more complicated since we have both an ideal and actual stream. The method to find enthalpy, entropy, quality, and temperature are given in previous sections. The temperature will differ between an ideal and an actual process, so it stands to reason that the pressure will also differ. However, the ideal pressure of the turbine does not have a clear

way to calculate, thus a guess must be made. I guessed $P_{6(\text{ideal})}=5$ bar. I made this guess because temperature increases compared to the irreversible process, thus pressure must also increase. For now, the topic of ideal pressure will be shelved. To find work, since it is not explicitly given for either the ideal or actual stream, we must calculate each using the energy balance. The work is as follows:

$$W_{s(\text{turbine})} = m\Delta H, \text{ where } W_{s(\text{turb})} < 0 \text{ because turbines produce power}$$

This function holds true for both the ideal and actual work. After finding the power for both streams, using the guessed pressure, we will use the efficiency to find the true pressure for the ideal process. The efficiency of the turbine as a function of ideal and actual work can be written as:

$$\eta = \frac{W_{s(\text{ideal})}}{W_s}, \text{ where } \text{abs}(W_{s(\text{ideal})}) > \text{abs}(W_s)$$

Next, I will use goal seek to set the efficiency of the turbine to the given efficiency, 0.85, while changing the pressure of the ideal turbine output (stream 6 ideal). This change in pressure only affects the ideal turbine output, so the energy balances for systems A-C remain unaffected. In other words, this ideal stream has no effect on the lost work or power requirement of any system.

Since the power consumed or produced from each equipment has been calculated, the power requirement of the entire system, in kW, can be found:

$$W_{S(\text{total})} = W_{s(\text{comp1})} + W_{s(\text{comp2})} + W_{s(\text{comp3})} + W_{s(\text{comp4})} + W_{s(\text{turbine})}$$

I then convert from kW to MW by dividing the total work by 1000. Compared to the Linde Process example done in the lecture, the power requirement is nearly three times lower. This shows that the four-stage compressor and turbine had a large effect on the power requirement.

Calculate Lost Work

The formula for calculating lost work can be found in the “Formulas and Balances” section of this report. For all systems excluding the compressors and the turbine, lost work will simply be the change in the availability flow, where $LW = -\Delta mB$. On the other hand, for the systems that involve work, $LW = -\Delta mB + W_s$. The formula for calculating availability can be found in the same section, and availability flow will be similar, only differing in how it uses enthalpy flow and entropy flow instead of enthalpy and entropy. The total lost work can be found by adding the lost work of each equipment, and can be written, using the same terminology in my diagram, as follows:

$$LW_{\text{total}} = LW_{\text{mixer}} + LW_{\text{CM1}} + LW_{\text{CL1}} + LW_{\text{CM2}} + LW_{\text{CL2}} + LW_{\text{CM3}} + LW_{\text{CL3}} + LW_{\text{CM4}} + LW_{\text{CL4}} + LW_{\text{HE}} + LW_{\text{TB}} + LW_{\text{DR}}$$

The percentage of lost work for each equipment is found by dividing the lost work of that individual equipment by the total lost work, and then multiplying by 100.

Identify the Goal

The goal is equal to the change in the availability flow of the overall system. The overall system, in our case, only has one input (stream 1) and one output (stream 7). The goal is as follows:

$$Goal = m_1 B_1 - m_7 B_7 = m_1 (H_1 - T_0 S_1) - m_7 (H_7 - T_0 S_7)$$

Calculate Thermodynamic Efficiency

The goal and total lost work have been found, thus allowing thermodynamic efficiency to finally be calculated. The goal is negative, so we use the following formula:

$$\eta = \frac{+Goal}{Goal - (LW)}$$

As can be seen in the second image on the next page, the thermodynamic efficiency is 9.33%, which is over double that of the Linde Process we did in the lecture, which had a thermodynamic efficiency of 4.03%.

Conclusion

All desired calculations, including power consumption, lost work in every equipment, total lost work, and thermodynamic efficiency, have been completed. The total power requirement and thermodynamic efficiency show substantial improvements, indicating that the four-stage compressor and turbine are effective. However, the enhancements were not as significant as expected. I likely encountered problems with the ideal turbine stream, possibly selecting an incorrect thermal property to guess or misapplying the lever rule. It's also possible that the turbine efficiency was inaccurately determined. Despite exploring several methods, most did not yield satisfying results or required additional assumptions. Ultimately, I opted to use the "GasTurbTemp()" function for temperature, which allowed me to use the lever rule for quality and guess pressure. This limited the ideal turbine stream to one equation (turbine efficiency) and one unknown (pressure), thus allowing me to use goal seek without disrupting any other balances.

ENG Calculations	Peng-Robinson															
Stream Properties	S1	S2	SCM1	SCL1	SCM2	SCL2	SCM3	SCL3	S3	S4	S5	S6(ideal)	S6	S7	S8	S9
Molar Flow, kmol/s	0.6917	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	10.7897	0.6917	10.0980	10.0980
Mass Flow, kg/s																
Volume Flow, m ³ /s																
Temperature, C	20.0000	24.8794	181.2457	30.0000	189.4910	30.0000	190.0610	30.0000	190.5092	30.0000	-30.3146	-162.2799	-195.9109	-195.9109	-195.9109	25.0000
Pressure, bar	1.0000	1.0000	3.6628	3.6628	13.4164	13.4164	49.1422	49.1422	180.0000	180.0000	180.0000	8.8913	1.0000	1.0000	1.0000	1.0000
Vapor Fraction	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.8805	0.9359	0.0000	1.0000	1.0000
Molecular Weight, kg/kmol																
Molar Enthalpy, kJ/kmol	-153.844	-17.043	4560.351	118.863	4782.036	48.512	4729.183	-188.709	4591.126	-786.336	-3100.269	-6371.553	-6948.838	-12153.363	-6592.318	-7.672
Vapor Phase	-153.844	-17.043	4560.351	118.863	4782.036	48.512	4729.183	-188.709	4591.126	-786.336	-3100.269	-5893.020	-6592.318	0.000	-6592.318	-7.672
Liquid Phase	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	-9897.700	-12153.363	-12153.363	0.000	0.000
Molar Entropy, kJ/(kmol K)	-0.407	0.056	1.607	-10.277	-8.725	-21.273	-19.718	-32.763	-31.178	-45.473	-54.005	-54.005	-45.087	-112.469	-40.471	0.087
Vapor Phase	-0.407	0.056	1.607	-10.277	-8.725	-21.273	-19.718	-32.763	-31.178	-45.473	-54.005	-49.274	-40.471	0.000	-40.471	0.087
Liquid Phase	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	-88.867	-112.469	-112.469	0.000	0.000
Molar Density, kmol/m ³																
Enthalpy Flow, kJ/s	-106.42	-183.89	49204.78	1282.49	51596.68	523.43	51026.42	-2036.11	49536.82	-8484.32	-33450.94	-68747.08	-74975.81	-8406.85	-66568.96	-77.47
Entropy Flow, kJ/(s K)	-0.28	0.60	17.34	-110.88	-94.14	-229.53	-212.75	-353.50	-336.40	-490.64	-582.70	-582.70	-486.47	-77.80	-408.67	0.88
Availability, kJ/s	-21.95	-365.01	44002.73	34547.40	79838.64	69382.87	114852.81	104014.18	150457.30	138707.29	141359.02	106062.88	70965.59	14932.55	56033.04	-342.24
Duty (Power), kJ/s [kW]			49388.67		50314.19		50502.99		51572.93			-35296.14	-41524.87			
Efficiency			0.85		0.85		0.85		0.85			0.85				
Compressibility Factor, Vapor	0.9995	0.9996	1.0009	0.9986	1.0037	0.9953	1.0147	0.9884	1.0670	1.0204	0.9512	0.8482	0.9593		0.9593	0.9996
Compressibility Factor, Liquid												0.0420	0.0048	0.0048		

ENG Calculations		
User Data and Calculations		
Energy Balance (System A)	0.0000	
Energy Balance (System B)	0.0000	
Energy Balance (System C)	0.0000	
Power Requirement, MW	160.2539	
Goal, KW	-14954.4998	
Lost Work (MIXER), kW	0.8080	0.00%
Lost Work (Compressor 1), kW	5020.9288	3.46%
Lost Work (Compressor 2), kW	5022.9520	3.46%
Lost Work (Compressor 3), kW	5033.0496	3.46%
Lost Work (Compressor 4), kW	5129.8174	3.53%
Lost Work (Cooler 1), kW	9455.3277	6.51%
Lost Work (Cooler 2), kW	10455.7776	7.20%
Lost Work (Cooler 3), kW	10838.6216	7.46%
Lost Work (Cooler 4), kW	11750.0110	8.09%
Lost Work (HEX), kW	53723.5538	36.97%
Lost Work (Turbine), kW	28868.5652	19.87%
Lost Work (DRUM), kW	0.0000	0.00%
Total Lost Work, kW	145299.4128	
Thermodynamic Efficiency	9.33%	

1 Introduction

The Linde process is a common method for gas liquefaction that operates by compressing and cooling the gas before expanding it to induce a phase change. In class, we looked at the Linde process for nitrogen liquefaction, which uses a single-stage compressor and an expansion valve. While effective, this configuration has a low thermodynamic efficiency, which was calculated to be 4%, thus contributing to high energy consumption and increased operating costs.

To improve thermodynamic efficiency, we are going to replace the single-stage compressor with a four-stage compressor with intercoolers in between them to bring the discharge down to 30 °C. We are also going to substitute the expansion valve with a liquid turbine (expander).

The objective of this assignment is to look at the impact of these modifications on the overall thermodynamic efficiency of the Linde process. Using the Peng-Robinson equation of state to describe the thermodynamic properties of nitrogen, we will analyze the modified process under the following assumptions:

- The efficiency of the compressor and liquid turbine (expander) is 85%.
- There is no pressure drop across the separator drum, intercoolers, and heat exchanger.

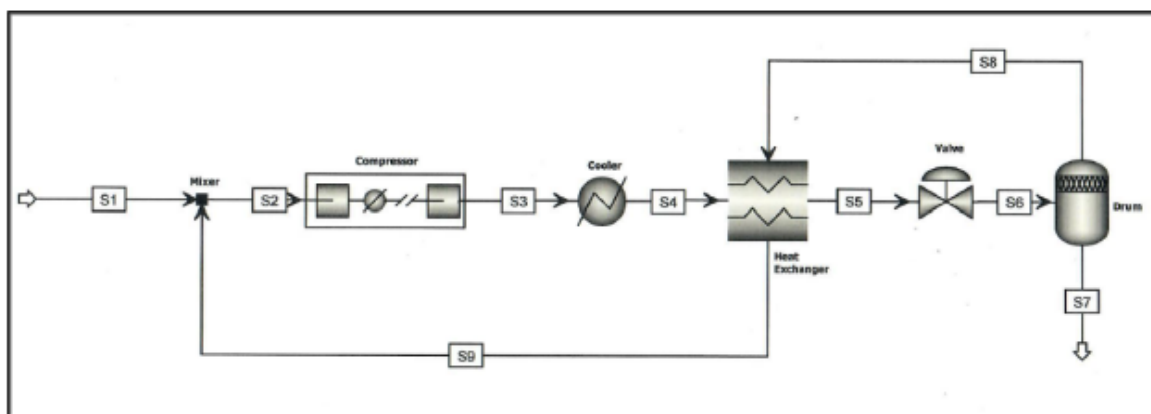


Figure 1: Process flow diagram for the Linde process

Stream No →	S1	S2	S3	S4	S5	S6	S7	S8	S9
Temperature, °C	20.0			30.0		T^{sat}	T^{sat}	T^{sat}	25.0
Pressure, bar	1.0	1.0	180.0	180.0	180.0	1.0	1.0	1.0	1.0
Flow Rate, MMSCFD	50.0								

Figure 2: Given Data

2 Stream Properties

2.1 Energy Balance

$$-\Delta mH + Q + W_s = 0 \quad (1)$$

2.2 Lost Work Balance

$$LW = -\Delta m(H - T_0S) + Q \left(1 - \frac{T_0}{T}\right) + W_s \quad (2)$$

2.3 Goal: Change in the Availability

Availability $B = H - T_0S$, where $T_0 = 300$ K.

2.4 Breaking Down the Process

Using the CHE Calculator and activating the Peng-Robinson equation of state and thermodynamic properties, we can set up our calculations in the calculator.

Since it is a four-stage compressor, we can label the stream inside the compressors to be S2Cn, where n is the corresponding stream number.

2.5 Molar Flow (kmol/s)

- **S1:** To simplify our calculations, we can convert the given S1 flow rate from MMSCFD using the CHE Calculator function =MMSCFDtoKMOLSEC().
- **S7:** Since it is a closed steady-state system, the molar flow of S1 and S7 are the same.
- **S9:** To calculate the flow rates for the rest of the streams, we can guess the flow rate of S9 to be 5 kmol/s. We will use goal seek later to figure out the actual flow rate of S9, which will give us the appropriate flow rate for all streams.
- **S2:** Since the stream S9 is being added to the mixer, $M_{S2} = M_{S1} + M_{S9}$.
- **S2C1 to S6:** Since the flow rate does not change due to the closed and steady-state nature of the process, the molar flow rate is the same as S2.
- **S8:** Since stream S6 is being divided into S8 and S7, $M_{S8} = M_{S6} - M_{S7}$.

2.6 Pressure (bar)

- **S2C1 to S2C6:**
 - The initial pressure at S1 is 1 bar, and the final discharge pressure at S5 is 180 bar.
 - The overall compression ratio is:
$$\text{Total compression ratio} = \frac{\text{final pressure}}{\text{initial pressure}} = \frac{180}{1} = 180$$
 - Since there are four stages, we assume the pressure increases by the same ratio at each stage. The per-stage compression ratio would be the fourth root of the total compression ratio:

$$\text{Per-Stage Compression Ratio} = 180^{1/4} = 3.66$$

Starting with an initial pressure of 1 bar at S1, the pressure after each compressor can be calculated as follows:

- After Compressor 1 (S2C1): $1 \text{ bar} \times 3.66 = 3.66 \text{ bar}$
- After Compressor 2 (S2C3): $3.66 \text{ bar} \times 3.66 = 13.42 \text{ bar}$
- After Compressor 3 (S2C5): $13.42 \text{ bar} \times 3.66 = 49.14 \text{ bar}$
- After Compressor 4 (S3): $49.14 \text{ bar} \times 3.66 = 180 \text{ bar}$ (given)

Since the outlet pressure from the compressor does not change after it goes through the intercooler, the outlet pressure of the compressor is the same as the inlet and outlet pressure of the intercooler.

2.7 Temperature (°C)

After inputting the rest of the given data into the calculator, we can use the `=EOSTsat()` function to calculate the saturated temperature using the given pressure. If we guess the temperature of S2, then we can calculate the temperature of S2C1 to S3. Since $T_{S1} = 20^\circ\text{C}$ and $T_{S9} = 25^\circ\text{C}$, then T_{S2} is somewhere in between, so we assume $T_{S2} = 21^\circ\text{C}$.

To calculate the rest of the properties, we could also guess the temperature of S5 to be -100°C .

- **S2C1 to S3:** We can use the function `=GasCompTemp()` to calculate the outlet temperature of each compressor using the inlet pressure, inlet temperature, outlet pressure, and compressor efficiency, which is 0.85.

2.8 Vapor Fraction

- **S1 to S5:** The vapor fraction is 1 since N_2 is in the gas phase.
- **S6:** Since the stream is a vapor-liquid mixture, it is less than 1. We can calculate the vapor fraction by dividing the molar flow of S8 by S6.
- **S7:** Since it is N_2 in its liquid state, $V_f = 0$.
- **S8:** Since it is the vapor from the drum, $V_f = 1$.
- **S9:** Since it is the recycle vapor stream, $V_f = 1$.

Now that we know each stream's temperature, pressure, molar flow, and vapor fraction, we can use the Peng-Robinson EOS to calculate all thermal properties.

3 Thermal Properties

3.1 Compressibility Factor

We can calculate the vapor compressibility factor for each stream, except for S7, by using the function `=zcomp()`, which takes temperature, pressure, and an initial guess for the compressibility factor as inputs. In our case, the initial guess for the compressibility factor would be 1 (for the liquid phase).

For S7 and S6 (a vapor-liquid mixture), we use the same function with the same variables, but with an initial guess for the compressibility factor set to 0, as they are in the vapor phase.

3.2 Enthalpy (kJ/kmol)

- **Vapor Phase:** We can calculate the vapor phase enthalpy for all streams, except S7, using the function `=Enthalpy()`, which requires temperature, pressure, and compressibility factor as inputs. Since S7 is in the liquid phase, we can set its vapor phase enthalpy to 0.
- **Liquid Phase:** Since all streams except for S6 and S7 are in the vapor phase, we can set the liquid phase enthalpy to 0 for these streams. For S6 and S7, we use the `=enthalpy()` function again, but this time with the vapor compressibility factor.
- **Molar Enthalpy:** We can calculate the molar enthalpy by applying the lever rule for enthalpy:

$$H = x \cdot H_v + H_l \cdot (1 - x)$$

where H is the molar enthalpy, x is the vapor fraction, H_v is the vapor phase enthalpy, and H_l is the liquid phase enthalpy.

3.3 Entropy (kJ/(kmol K))

- **Vapor Phase:** We can calculate the vapor phase entropy for all streams, except for S7, using the function `=Entropy()`, which takes temperature, pressure, and compressibility factor as inputs. Since S7 is in the liquid phase, we set its vapor phase entropy to 0.
- **Liquid Phase:** Since all streams except for S6 and S7 are in the vapor phase, we set the liquid phase entropy to 0 for these streams. For S6 and S7, we use the `=entropy()` function again, but with the vapor compressibility factor.
- **Molar Entropy:** Using the lever rule for entropy, we can calculate the molar entropy as:

$$S = x \cdot S_v + S_l \cdot (1 - x)$$

where S is the molar entropy, x is the vapor fraction, S_v is the vapor phase entropy, and S_l is the liquid phase entropy.

3.4 Enthalpy Flow (kJ/s)

The enthalpy flow can be calculated by multiplying the molar enthalpy by the molar flow rate.

3.5 Entropy Flow (kJ/(s K))

The entropy flow can be calculated by multiplying the molar entropy by the molar flow rate.

3.6 Availability (kJ/s)

The availability can be calculated by subtracting the enthalpy flow from the product of the reference state temperature (300 K) and the entropy flow.

3.7 Duty/Power (kW)

In our process, power calculations are relevant for each compressor and the liquid turbine. Power can be calculated by taking the difference in enthalpy flow between the inlet and the outlet.

4 User Data and Calculations

Currently, we have three unknowns: T_{S2} , T_{S5} , and F_{S9} . We can find these unknowns using the energy balance equations, Goal Seek, and the Newton method.

We can divide the process into three systems:

- **System A:** Mixer
- **System B:** Heat exchanger, liquid turbine, and drum
- **System C:** Liquid turbine

4.1 System A

For System A, we use the following energy balance:

$$-\Delta(mH) + Q + W_s = 0 \quad (3)$$

Since $\Delta(mH) = 0$, this simplifies to:

$$m_1 H_1 + m_9 H_9 = m_2 H_2$$

Using Goal Seek, we can set the equation to 0 by adjusting the cell containing the guessed temperature value in S2. This will yield the correct temperature for T_{S2} .

4.2 System B

For System B, we use the Newton method to set both System B and System C to equal 0, while adjusting the values for F_{S9} and T_{S2} .

4.3 System C

In System C, we apply the Newton method again, but this time we set all three systems to equal 0 by adjusting all the unknown variables: T_{S2} , T_{S5} , and F_{S9} . This provides the correct values for each unknown.

The resulting values are:

$$\begin{aligned}T_{S2} &= 24.68^{\circ}\text{C} \\T_{S5} &= -30.31^{\circ}\text{C} \\F_{S9} &= 10.1 \text{ kmol/s}\end{aligned}$$

4.4 Lost Work (kW)

- **Mixer (kW):** We can calculate the lost work for the mixer by adding the availability of the inlet streams, S1 and S9, and subtracting the availability of the outlet stream S2.
- **Turbine, Compressors 1, 2, 3, and 4:** Lost work is calculated by subtracting the availability of the inlet and outlet streams and adding the calculated power value.
- **Cooler, Intercoolers 1, 2, and 3:** Lost work is determined by subtracting the availability of the inlet and outlet streams.
- **Heat Exchanger:** Lost work is calculated by subtracting the availability of the inlet streams (S4 and S8) from the availability of the outlet streams (S5 and S9).
- **Drum:** Lost work is determined by subtracting the availability of the inlet stream S6 and the outlet streams S7 and S8.

The total lost work is the sum of all the lost work values calculated above:

$$\text{Total Lost Work} = 145299.41 \text{ kW}$$

The lost work percentage for each component can be calculated by dividing the lost work from one component by the total lost work, then multiplying by 100.

4.5 Thermodynamic Efficiency (%)

- **Goal (kW):** The change in availability from S1 to S7 is -14954.5 kW .
- **Thermodynamic Efficiency:**

$$\text{Thermodynamic Efficiency} = \frac{\text{Goal}}{\text{Goal} - \text{Total Lost Work}} = 9.33\%$$

5 Conclusion

In the Linde process with a single-stage compressor and throttling valve, we obtained a thermodynamic efficiency of 4.03%. However, after replacing the single-stage compressor with a four-stage compressor and substituting the valve with a liquid turbine, our efficiency improved by more than 5.3 percentage points, reaching 9.33%.

Thus, we were able to more than double the efficiency with the modified process. This improvement is attributed to reduced irreversibility from intercooling between compression stages and the work recovery achieved through the liquid turbine, which significantly lowered the overall energy consumption.